

MODULE - I

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22/10/24

Classification of data, Characteristics, Evolution & definition of Big data, What is Big data, Why Big data. Traditional Business Intelligence Vs Big Data, Typical data warehouse and Hadoop environment.

Big Data Analytics: What is Big data Analytics, classification of Analytics, Importance of Big Data Analytics, Technologies used in Big data Environments, Few Top Analytical Tools, NoSQL, Hadoop.

1.1 CLASSIFICATION OF DIGITAL DATA:

Data $\rightarrow$ information

Information $\rightarrow$ Insights [An accurate & deep understanding]

Digital data - It is information that is represented as a series of Symbols & is shared electronically.

Digital data can be broadly classified into 3 types

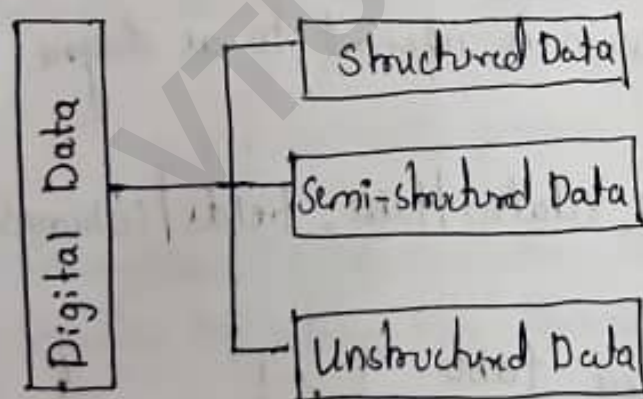


Fig: Classification of digital data.

1. Structured Data:

$\rightarrow$ This is the data which is in an organized form & can be easily used by a computer program.

$\rightarrow$ Ex: RDBMS i.e. (Data stored in rows & columns format), Relationship exist b/w entities of data, such as classes & their objects. Data stored in database is also in structured format.

→ When do we say data is structured?

Ans:- When data conforms to a pre-defined schema/structure we say it is structured data.

→ Most of the structured data is held in RDBMS.

→ RDBMS conforms to the relational data model wherein the data is stored in rows/columns

Ex: College Database

Student Details

Student-id	Stud-name	Stud-Age	Stud-DOB	Stud-MobileNo
1	Rob	21	21/05/99	9123456710
2	Joy	22	22/07/2000	2105432618
3	Riya	23	25/08/1998	9632487610

→ The no. of records in a relation is called as Cardinality of a relation & the no. of column is referred to as degree of a relation

→ First step is design of a relation/table, fields/columns to the store the data

Ex: Sample records in Employee table.

EmpNo	EmpName	Designation	DeptNo	ContactNo
E101	Allen	Software Engineer	D1	0999999999
E102	Simon	Consultant	D1	0777777777

Sources of structured data

[Various places or systems where data is stored in well-defined format] (3)

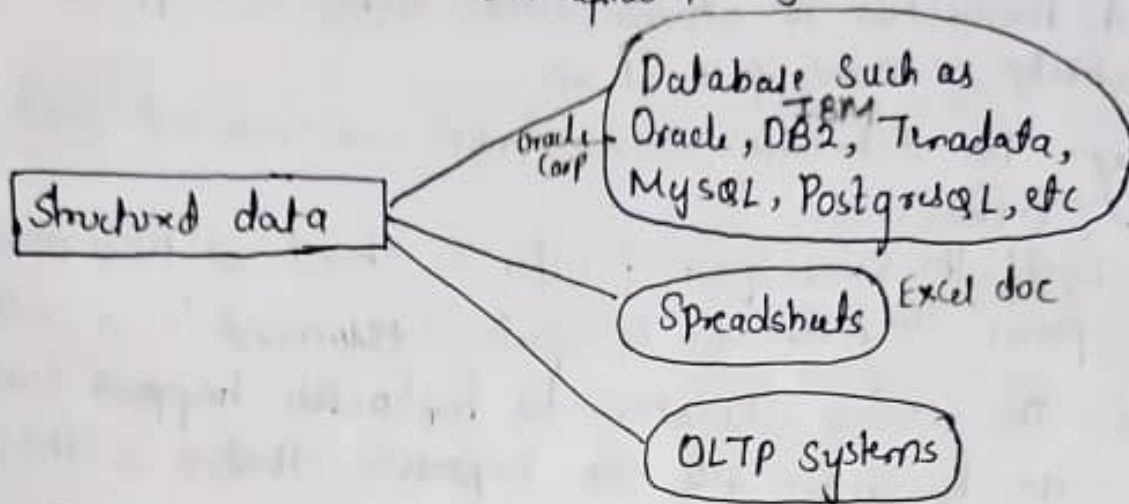


Fig: Sources of structured data.

→ These databases are typically used to hold transaction/operational data generated & collected by day-to-day business activities.

→ In other words, the data of the online Transaction processing (OLTP) systems are generally quite structured.
Ex: Banking activities

Ease Of Working with Structured Data:

1. Insert/Update/Delete: The DML [Data Manipulation language] operation provide the required ease with data i/p, storage, access, process, analysis etc.

2. Security: Encryption & tokenization solution to warrant the security of information throughout its lifecycle.

3. Indexing: It is a data structure that speeds up the data retrieval operations at the cost of additional writes & storage space.

4. Scalability: The traditional RDBMS storage & processing capabilities can be easily scaled up by increasing horsepower of database server.

5. Transaction processing: RDBMS supports ACID properties
- Acid: - A transaction is atomic, means either it happens in its entirety or none of it at all.
- Consistency: The database moves from one consistent state to other.
i.e. If the same piece of info is stored at two or more places, they are in complete agreement.
- Isolation: The resource allocation to transaction happens such that the transaction gets the impression that it is the only transaction happening in isolation.
- Durability: All changes made to the database during a transaction are permanent & the accounts for the durability of the transaction.

~~2. Semi-structured Data:~~

~~It is also referred to as self-describing structure.~~

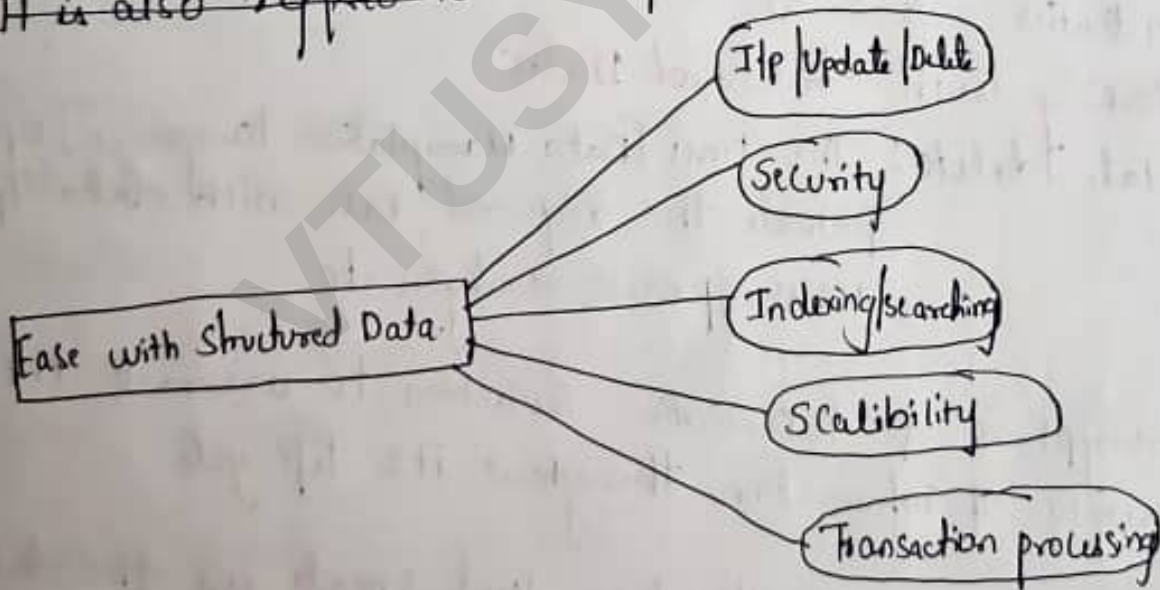


Fig: Ease of working with structured data.

2. Semi-structured Data:

It is also referred to as self-describing structure.

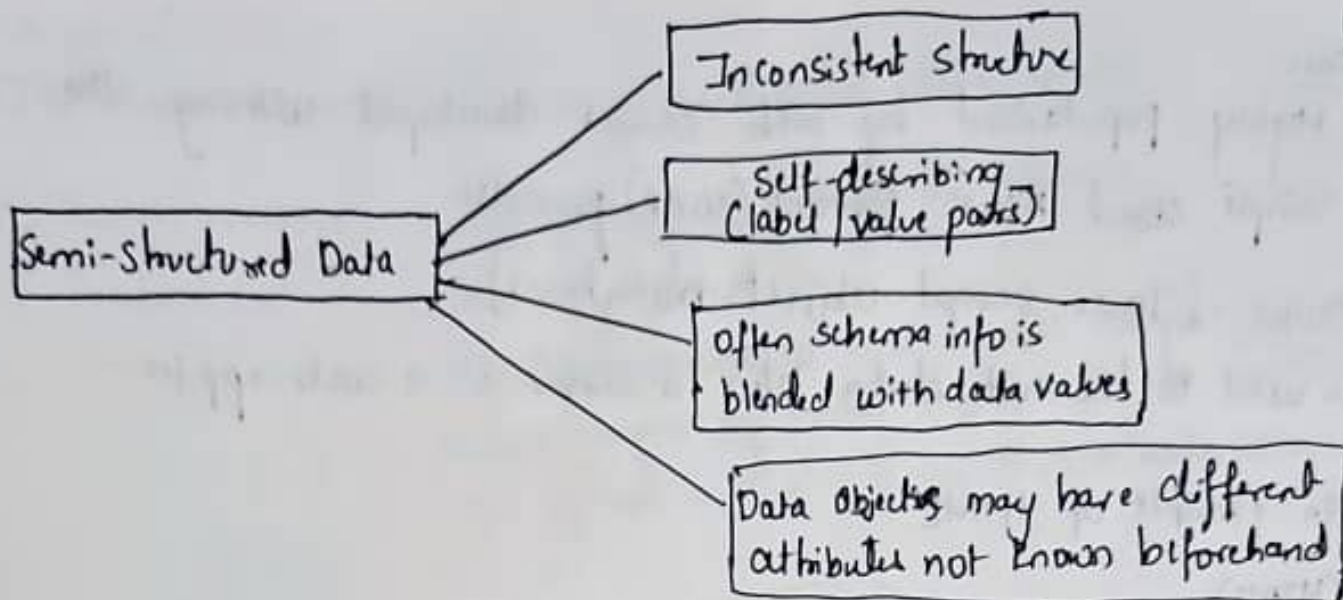


Fig: Characteristics of Semi-Structured Data.

Semi-structured data has the following features

1. It does not conform to the data models that one typically associates with relational databases or any other form of data tables
2. It uses tags to segregate semantic elements
3. Tags are also used to enforce hierarchies of records & fields within data.
4. There is no separation b/w the data & the schema. The amount of structure used is dictated by the purpose at hand.
5. In semi-structured data, entities belonging to the same class & also grouped together need not necessarily have the same set of attributes. And if at all, they have the same set of attributes, the order of attributes may not be similar & for all practical purposes is not important as well.

Sources of Semi-structured Data

The sources of Semi-structured are 'XML' & 'JSON'

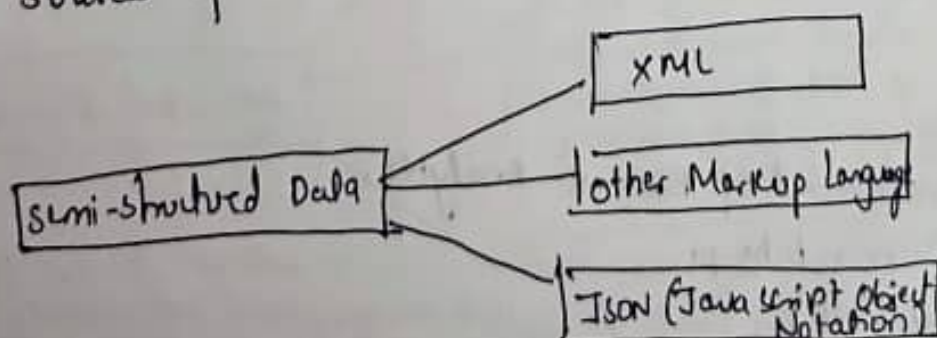


Fig: Sources of Semi-structured data

1. XML:

→ Highly popularized by web services developed utilizing the Simple Object Access Protocol (SOAP) principles.

2. JSON: [Java script object notation]

→ used to transmit data b/w a server & a web app^n.

An example of HTML

<HTML>

<HEAD>

<TITLE> Place your title here </TITLE>

</HEAD>

<BODY BGCOLOR="FFFFFF">

<CENTER> </CENTER>

<HR>

Link Name

<H1>this is a Header </H1>

<H2>this is a Sub Header </H2>

Send me mail at

Support@yourcompany.com

<p> a new paragraph!

<p> a new paragraph!

 <i> this is a new sentence without a paragraph break, in bold italics, </i>

<HR>

</BODY>

</HTML>

Sample JSON Ex:

{

id: 9;

BookTitle: "Fundamental of Business Analytics";

AuthorName: "Seema Acharya";

Publisher: "Wiley India";

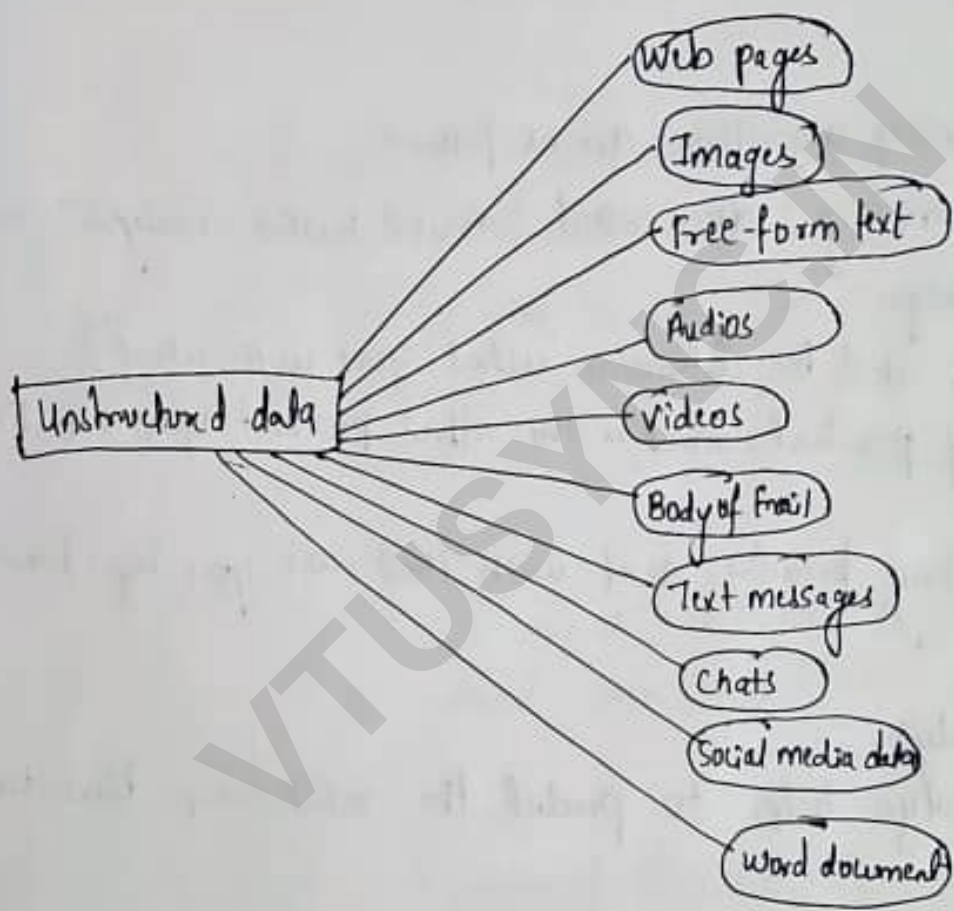
Year of publication: "2017";

3 Unstructured Data:

It does not conform to any pre-defined data model.

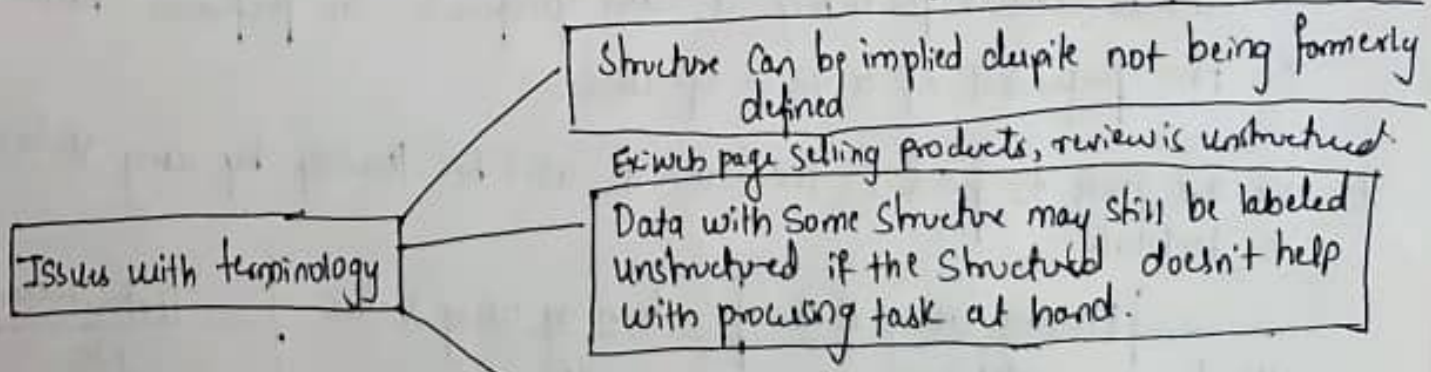
Ex: Twitter message
Facebook post
Email

Felling miffed 😞. Victim of twishing
LOL. cya. BFN
Hey Joan, possible to send across the first lot on Hadoop chapter by Friday Eop or may be we can meet up over a cup of coffee. Best regards, Tom



Issues with "Unstructured" Data

Ex: Review in the Restaurants



Ex: Think of tweets. They have a simple structure (text, hashtag, user mention) but there is a hidden layer of structure. Twitter's algo infers relationships b/w users, topics & communities.

Data may have some structure or may even be highly structured in ways that are unanticipated or unannounced

How to Deal with Unstructured Data

- Today unstructured data constitutes approximately 80% of data that is being generated in any enterprise
 - The data is clearly shifting to unstructured favor which is in such a big % which cannot be ignored.
 - There are few ways of dealing with unstructured data.
3. ~~Data mining~~ The ~~form~~ techniques are used to find patterns in or interpret unstructured data.

1. Data mining

Few data mining algorithms are as follows

- Association rule mining: also called "market basket analysis" or "affinity analysis"

→ It basically used to determine "what goes with what?"

→ If you buy product, what is the other product you are likely to buy

→ Ex: If you buy bread, next with that will you buy cheese or Egg

- Regression Analysis:

This algo helps to predict the relationship b/w two variables.

- Collaborative filtering:

It is about predicting a user preference or preferences based on the preferences of a group of users.

Ex: we are going to predict the user 4 will be learning by using videos or text book.

	Learning using Audios	Learning using Videos	Textual Learning
User 1	Yes	Yes	No
User 2	Yes	Yes	Yes
User 3	Yes	Yes	No
User 4	Yes	?	?

Text analytics or text mining:

- Compared to the structured data stored in a relational database, text is largely unstructured, amorphous, & difficult to deal with algorithmically
- It is the process of gaining high quality & meaningful info from text.

NLP:

- It is related to the area of human computer interaction.
- It is about enabling computers to understand human

Noisy text analytics:

- Process of extracting structured, semi-structured or ~~unstructured~~ info from noisy unstructured data such as chats, blogs, emails, text msgs etc
- The noisy unstructured data usually comprises one or more of the foll like Spelling mistakes, abbreviations, non-standard words, missing Punctuations, missing letters case, filler words such as "uh", "um" etc

5. Manual tagging with metadata:

Tagging manually with adequate metadata to provide the requisite semantics to understand unstructured data

6. Part of speech tagging

7. Unstructured Information Management Architecture (UIMA):

- Open source platform from IBM
- used for real-time content analytics.

2.1 CHARACTERISTICS OF DATA

Data has 3 key characteristics

1. Composition:

- Composition deals with structure of data i.e Source, granularity, types & the nature of data as to check whether it is static or real time streaming.

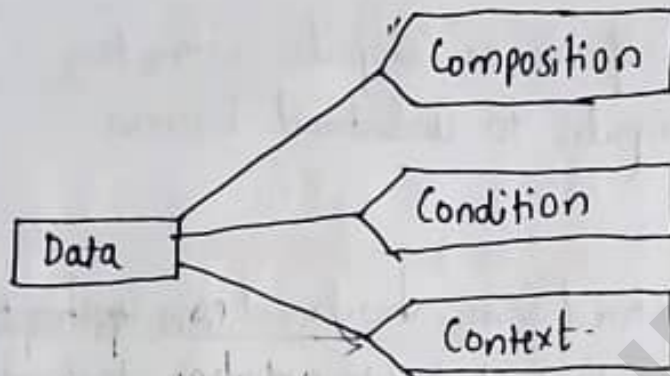
2. Condition:

- It deals with the state of data i.e "Can one use this data as is for

analysis?" or "Does it require cleaning for further enhancement & enrichment?"

3. Context:

It deals with "Where has this data been generated?" "Why was this data generated?" "How sensitive is this data?" "What are the events associated with this data?"



- 1) Is the data accurate & reliable?
- 2) Is data complete or are there any missing values?
- 3) Is data consistent?
- 4) Can the data be used as it is or any data needs to be cleaned?

Fig: Characteristics of data.

2.2 Evolution Of Big Data:

Data Generation & Storage		Data Utilization	Data Drivers
Complex & Unstructured			Structured data, Unstructured data, multistructured data.
Complex & Relational		Relational databases: Data-intensive applications	
Primitive & Structured	Mainframes: Basic data Storage 1970s & before	Relational (1980s & 1990s)	2000s & beyond

Table: The evolution of big data.

3 Definition of Big Data:

(11)

Defn:- Big data is high volume, high velocity, & high variety information assets that demand cost effective, innovative forms of info processing for enhanced insight & decision making

— Gartner IT Glossary

- The 3V's was proposed by Gartner analyst Doug Laney in a 2001 NutaGroup Search publication, titled, 3D Data Management: Controlling Data Volume, Variety & Velocity.
- For easy comprehension, we will look at the defin in three parts

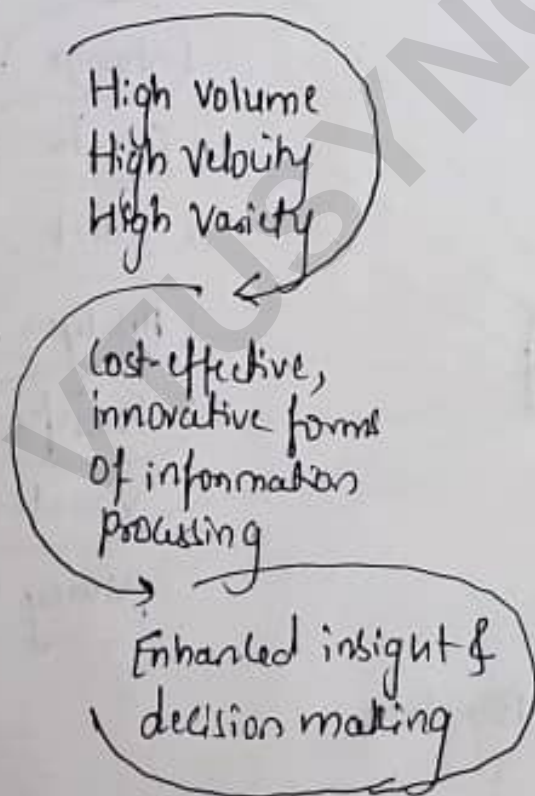


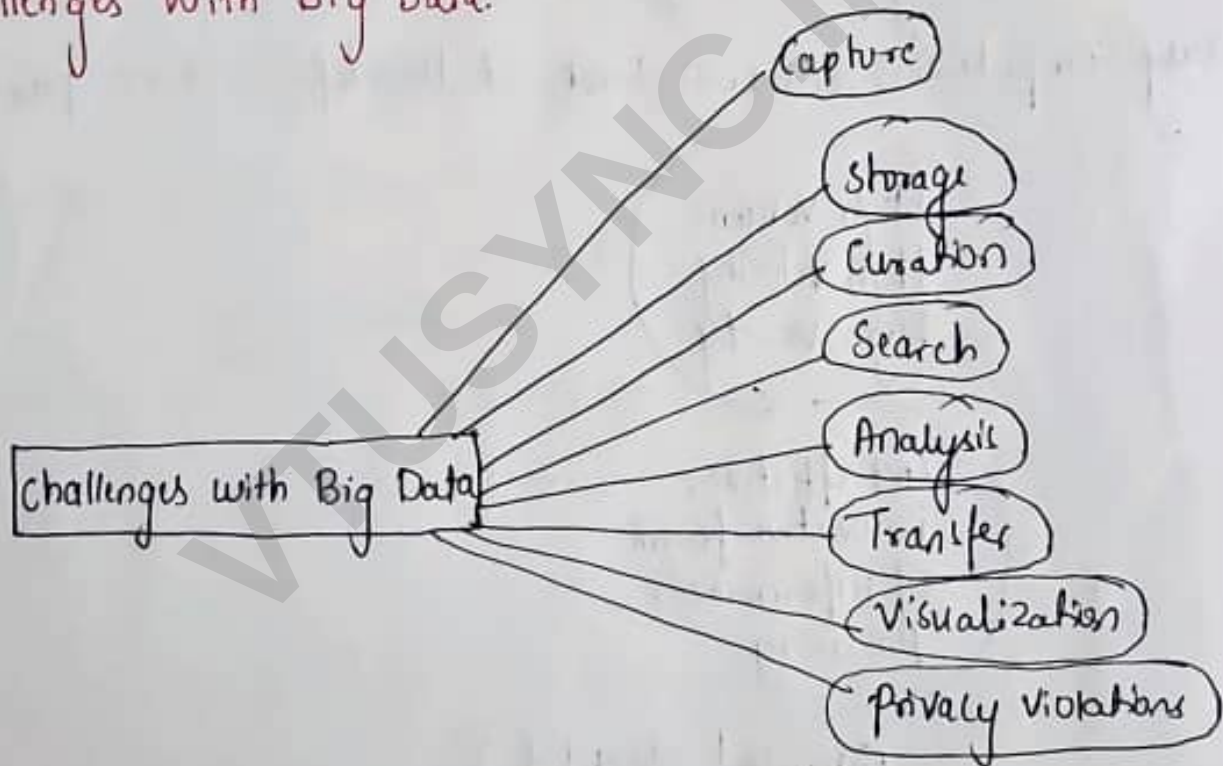
Fig: Definition of big data - Gartner.

- Part-I of the defin talks about Voluminous data that may have great variety (a good mix of structured, semi & unstructured data) & will require a good speed/space of storage, preparation, processing & analysis.

- Part II talks about embracing new techniques & technologies to Capture, Store, process, persist, integrate & visualize the high volume, high velocity, high variety data.
- Part III talks about deriving deeper, richer & meaningful insights & then using these insights to make faster & better decisions to gain business value & thus a competitive edge.

Data → Information → Actionable intelligence → Better decisions → Enhanced Business Value

2.4: Challenges With Big Data:



Pg: Challenges with Big data.

1) Explosive Data Growth:

We're creating massive amounts of data very quickly. The problem is figuring out what's useful & what's just noise. It's like trying to find a needle in a rapidly growing haystack.

2) Cloud Computing & Visualization

While CC offers cost-effective ways to manage big data, it also makes it harder to decide where to store & manage sensitive information.

3) Data Retention:

- Deciding how long to keep data is tricky.
- Some data is valuable for long-term decisions, while others become useless quickly.
- It's about figuring out what's worth keeping & what's not to avoid storing unnecessary information.

4) Lack of skilled Professionals:

- There aren't enough people with the right skills to effectively work with big data & create big data solutions.
- It's like having a lot of ingredients but not enough chefs to cook a great meal.

5) General Data Management Challenges:

- Big data is hard to capture, store, prepare, search, analyze, move, secure, & visualize.
- It's just massive & complex, making every step of the process difficult.
- The defn of "big data" is blurry but generally, it's data that's too big for normal database software to handle.

6) Data Visualization Expertise Shortage:

- Turning data into useful visuals (like charts & graphs) is a specialized skill.
- There aren't enough experts who can effectively present big data in a way that's easy to understand & use for decision making.

2.5: WHAT IS BIG DATA.

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→ It is data that is big in volume, velocity & variety

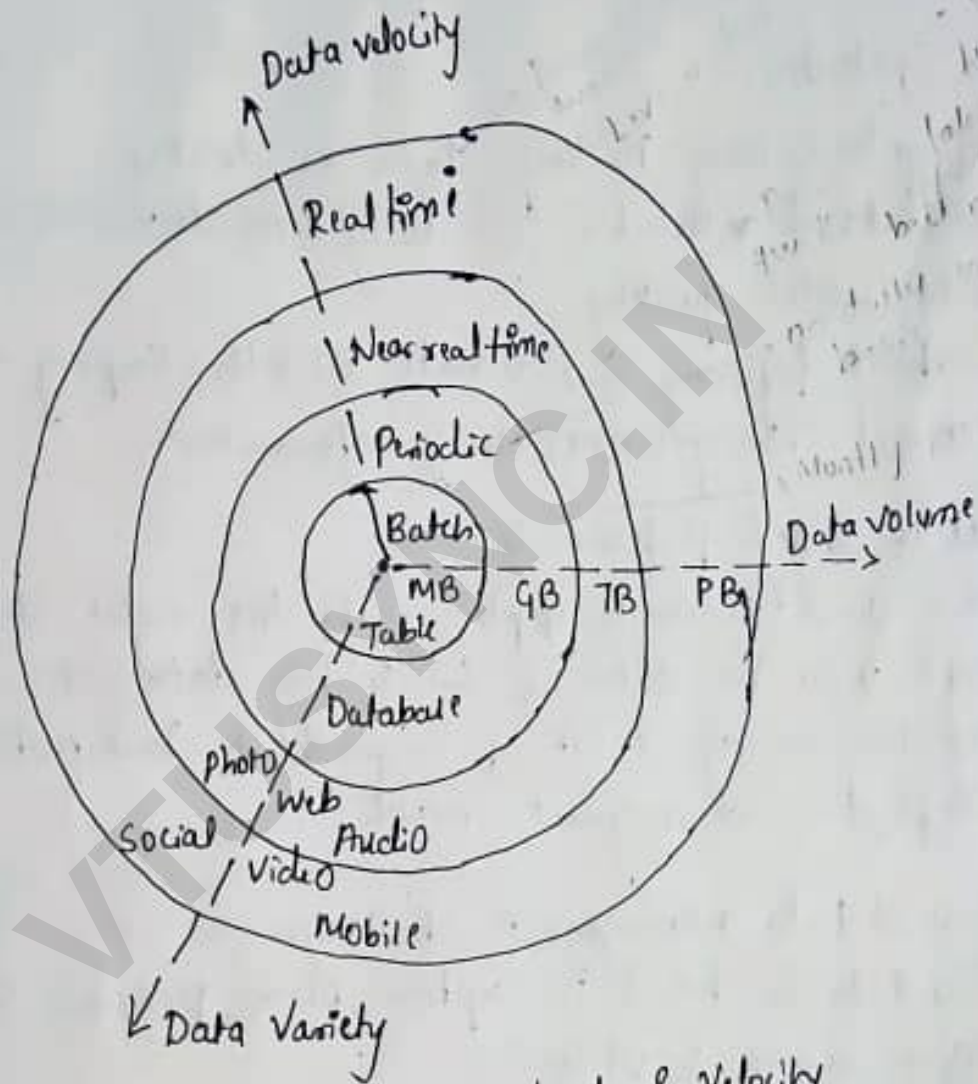


Fig: Data: Big in Volume, Variety & Velocity.

Volume:

It grows from bits to bytes to petabytes & exabytes.

Bits → Bytes → Kilobytes → Megabytes → Gigabytes → Terabytes
→ Petabytes → Exabytes → Zettabytes → Yottabytes

Table: Growth of data

Bits	0 or 1
Bytes	8 bits
Kilobytes	1024 bytes
Megabytes	1024^2 bytes
Gigabytes	1024^3 bytes
Terabytes	1024^4 bytes
Petabytes	1024^5 bytes
Exabytes	1024^6 bytes
Zetta bytes	1024^7 bytes
Yottabytes	1024^8 bytes

Where Does This Data gets Generated?

- An XLS, a Doc, a PDF etc.

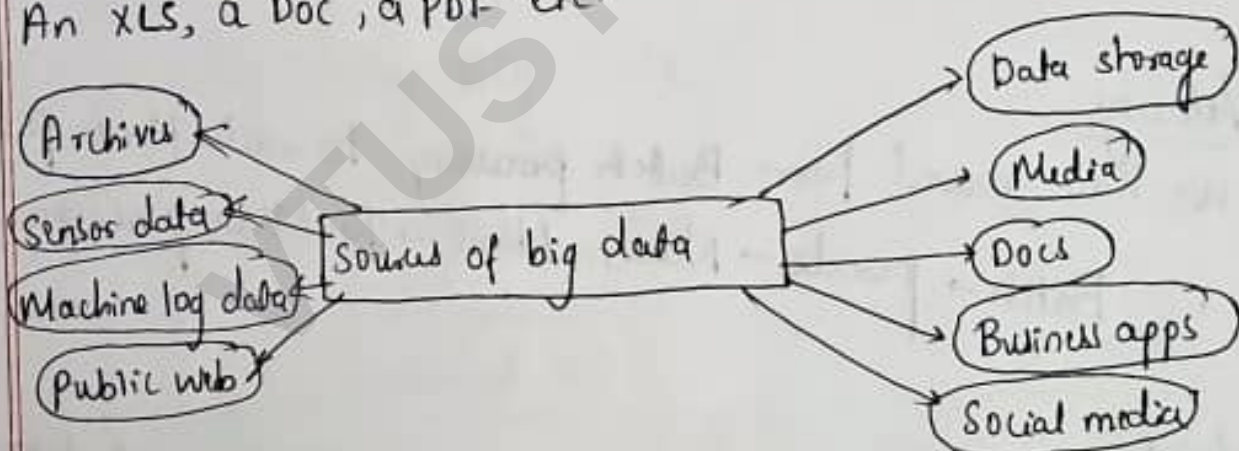


Fig: Sources of big data.

1. Typical internal data sources:
Data present within an organization's firewall
- Data storage: File systems, SQL (RDBMS - Oracle, MS SQL Server, DB2, MySQL, PostgreSQL, etc) NoSQL (MongoDB, Cassandra etc)
- Archives: Archives of scanned documents, paper archives, Customer Correspondence records, Patients health records, Students admission

records, Student assessment records

2. External data Source :

Data resides outside an organization's firewall

- Public web: Wikipedia, weather, regulatory, Compliance, Census etc.

3. Both (I + E)

- Sensor data
- Machine log data
- Social media
- Business app'n
- Media
- Doc.

Velocity:

We have moved from Batch processing to real-time processing

Batch → periodic → Near real time → Real time processing

Variety:

- It deals with wide range of data types & sources of data.

1. Structured
2. Semi-structured
3. Unstructured data.

2.7 WHY BIG DATA?

- The more data we have for analysis, the greater will be the analytical accuracy & greater would be the confidence in our decisions based on these analytical findings.

More data → More accurate analysis → Greater confidence in decision making → Greater operational efficiencies, Cost reduction, time reduction, new product development, & optimized offerings etc.

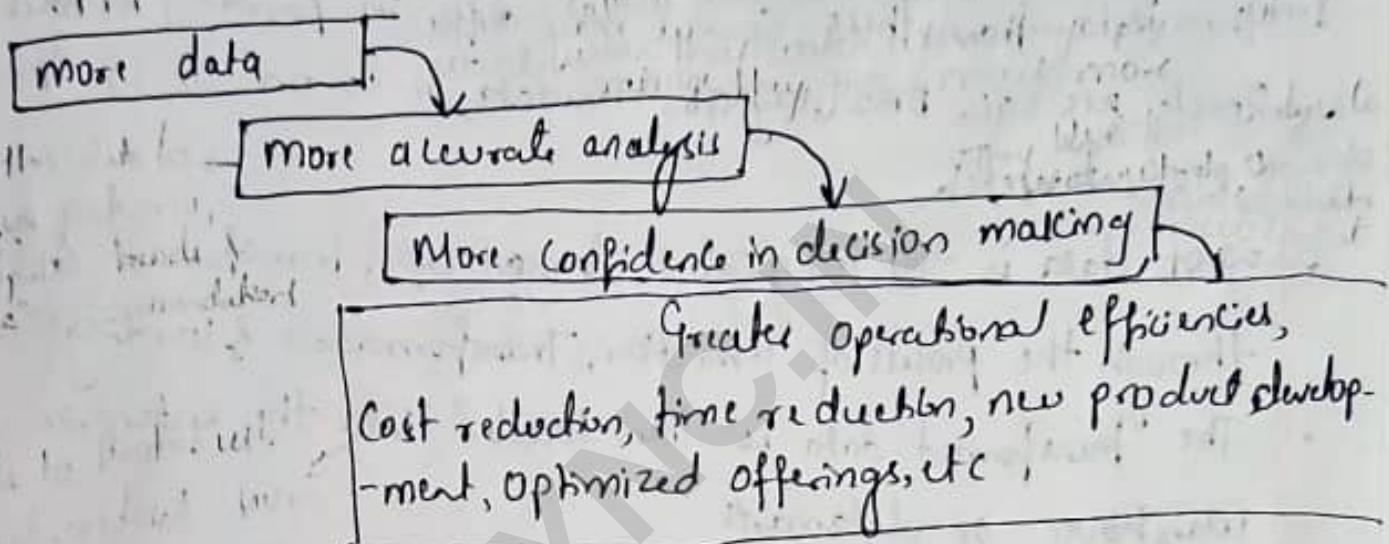


Fig: Why Big Data.

2.9 TRADITIONAL BUSINESS INTELLIGENCE (BI) VERSUS BIG DATA

	Traditional BI	Big Data.
Data Storage	Data is stored in Centralized server.	Data is stored in a distributed file system
Data Analysis	Data is analyzed offline (in batches).	Data can be analyzed both in real-time & offline
Data Type & Processing	Deals primarily with structured data & moves data to the processing function ("move data to code")	Deals with a variety of data types (structured, semi-structured, & unstructured) & moves the processing function to the data

2.10 A TYPICAL DATA WAREHOUSE ENVIRONMENT.

- Operational or transactional or day-to-day business data is gathered from Enterprise Resource planning (ERP), Customer Relationship Management (CRM), legacy systems, & several third party applications.
- The data from these sources may differ in format [RDBMS such as Oracle, MS SQL, DB2, MySQL, Teradata or so on or in Spreadsheet (.xls, .xlsx, etc)].
- This data is integrated, cleaned up, transformed & standardized through the process of Extraction, Transformation & loading (ETL)
- The transformed data is then loaded into the enterprise data warehouse or datamarts.
- A host of market leading business intelligence & analytics tools are then used to enable decision making from the use of ad-hoc queries, SQL, enterprise dashboards, data mining etc.

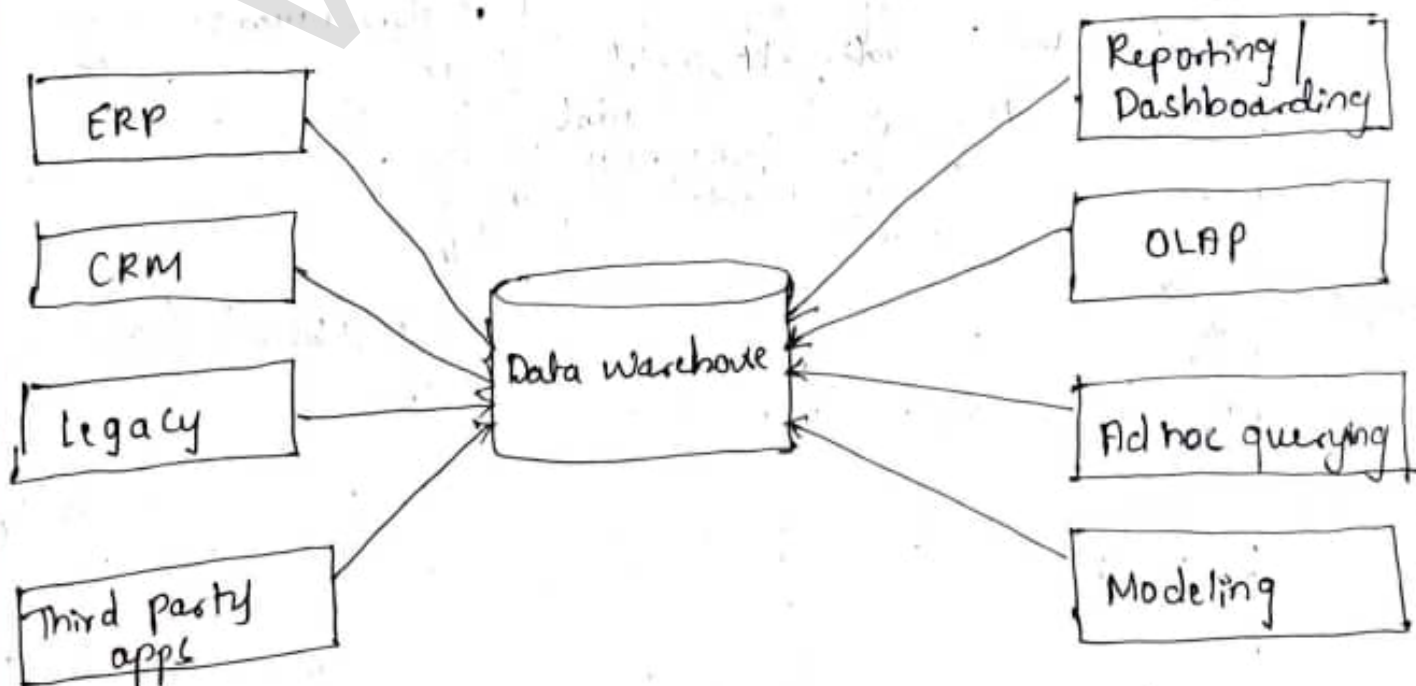


Fig: A typical data warehouse environment.

2.11 A Typical HADOOP ENVIRONMENT.

- In the hadoop environment the data sources are quite disparate from web logs to images, audios & videos to social media data to various docs, pdfs, etc.
- Here Data flows is not just the data within company's firewall but also residing outside the company's firewall
- The data is placed in HDFS.
- This can be repopulated back to operational systems or fed to the enterprise data warehouse or data marts or Operational Data stores (ODS) to be picked for further processing & analysis.

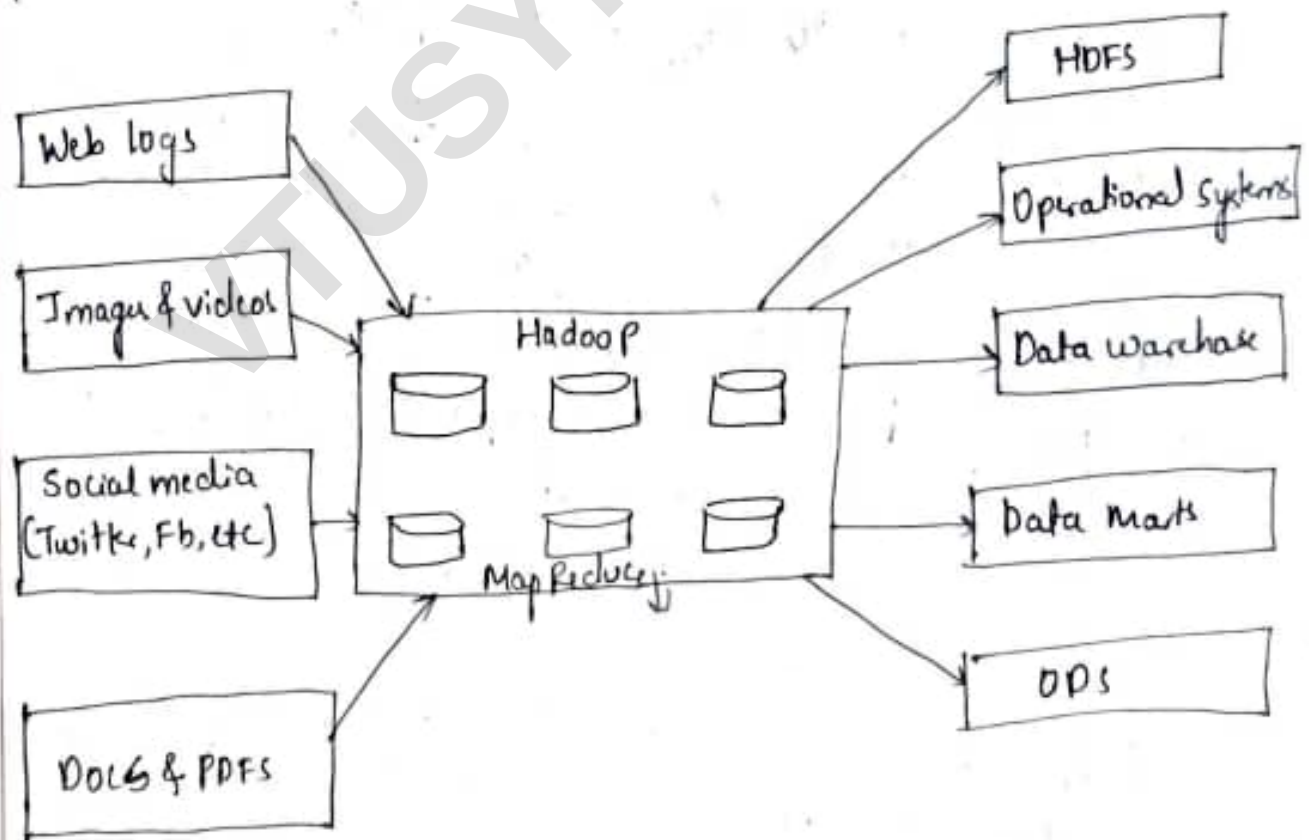


Fig: A Typical Hadoop environment.

BIG DATA ANALYTICS

Big Data analytics is:

- 1) Technology-enabled analytics:
 - Using Technology to understand lots of information
 - Its about using Special computer programs & tools to analyze huge amounts of data.
 - Tools such as IBM, Tableau, SAS, R Analytics, Statistica, World programming System(wps) etc. To help process & analyze your big data.
2. Finding important insights to make better decisions:
 - The goal is to discover hidden patterns & useful information in the data.
 - This helps business understand their customer better, improve their products & make smarter choices
3. Getting an edge over competitors:
 - By quickly analyzing data, companies can spot trends & opportunities faster than their rivals.
4. A tighter handshake b/w 3 communities IT, business & data scientists.
5. Working with data that's too big for regular computers:
 - It deals with massive amounts of data that traditional systems can't handle.
6. About moving code to data. This makes perfect sense as the program for distributed processing is tiny (just few KBs) compared to the data (Terabytes or petabytes today & likely

to be Exabytes or Zettabytes in the near future).

Ex: Imagine you buy clothes online. The website remembers what brands & colors you like. Later, it sends you personalized recommendations based on that information. That's big data analytics in action.

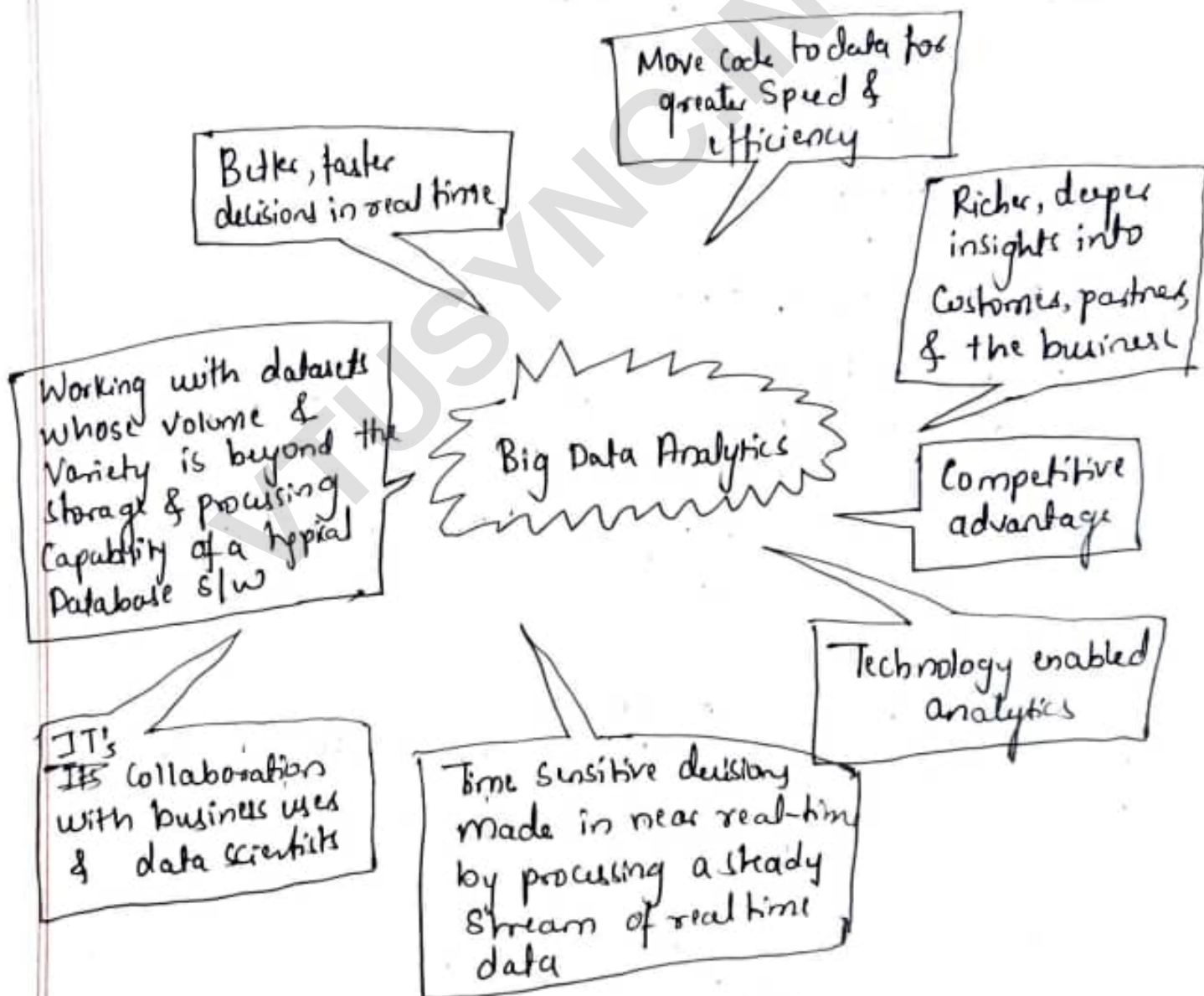


Fig: What is big data analytics.

3.5 CLASSIFICATION OF ANALYTICS

There are two schools of thought:

1) First school of thought

- Basic Analytics
- Operationalized Analytics
- Advanced Analytics
- Monetized Analytics

2) Second school of thought

- Analytics 1.0
- Analytics 2.0
- Analytics 3.0

1) First school of thought

- Basic Analytics
 - It is slicing & dicing of data to help with basic business insight.
 - This is about reporting historical data, basic, visualization, etc.

2) → Operationalized Analytics

- It is operationalized analytics if it gets woven into the enterprise's business processes.

→ Advanced Analytics.

- It is about forecasting for the future by way of predictive & prescriptive modelling.

→ Monetized Analytics

- This is analytics in use to drive direct business revenue.

What is Analytics

- It is the systematic computational analysis of data or statistics. It is used for the discovery, interpretation, & communication of meaningful patterns in data.
- It also entails applying data patterns towards effective decision making

2) Second school of thought

- Analytics 1.0
- Analytics 2.0
- Analytics 3.0

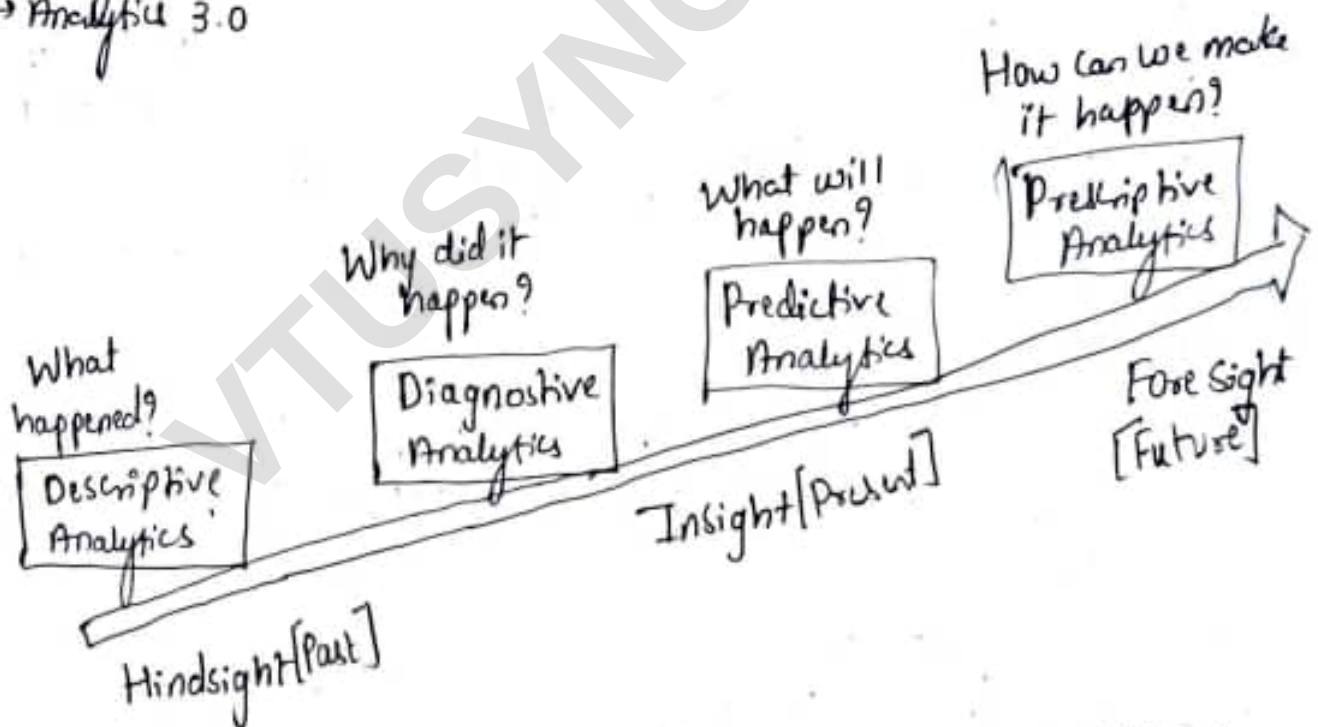


Fig: shows the steady growth of analytics from Descriptive → Diagnostic → Predictive → Prescriptive analytics

* Analytics 1.0

- Introduced in the era of mid 1950s to 2009
- Descriptive Statistics
- Data is taken from legacy systems, ERP, CRM etc.

- Data from structured data sources
- Data stored in enterprise data warehouse or data mart
- Data was internally sourced
- Relational Databases are used.

* Analytics 2.0

- Introduced in the era of 2005 to 2012
- Descriptive Statistics + predictive statistics
- Uses the past data
- Data is from unstructured sources
- Data was externally sourced
- Database apps, Hadoop clusters, SQL to Hadoop environment etc are used.

* Analytics 3.0

- Introduced in the era of 2012 to present
- Descriptive Statistics + predictive statistics + prescriptive statistics
- Uses the past data
- Data is blend of big data & data from legacy systems, ERP, CRM
- Data is internally & externally sourced.

3.8: Why is Big data Analytics important:

1) Reactive - Business Intelligence:

- What BI helps us with?
- It allows the business to make faster & better decisions by providing

- Right info to the right person at the right time in the right format.
 - It's about analysis of past or historical data & then displaying the findings of the analysis or report in the form of enterprise dashboards, alerts, notifications etc.
2. Reactive - Big Data Analytics:
 - Analysis is done on huge datasets but the approach is still reactive as it is still based on static data.
 3. Proactive - Analytics:
 - This supports futuristic decision making by the use of data mining, pre-dictive modeling, text mining & statistical analysis.
 4. Proactive - Big data Analytics:
 - This is sifting through terabytes, petabytes, exabytes of info to filter out the relevant data to analyze.
 - It includes high performance analytics to gain rapid insights from big data & the ability to solve complex problems using more data.

3.12 Terminologies used in Big Data Environments: i) In-memory Analytics

- Data access from non-volatile storage such as hard disk is a slow process.
- The more data is required to be fetched from the hard disk or secondary storage, the slower the process gets.
- One way to combat this challenge is to pre-process & store data so that the CPU has to fetch a small subset of records.
- This requires thinking in advance as to what data will be required for analysis.
- This problem has been addressed using in-memory analytics.
- Here all the relevant data is stored in RAM or primary storage.

thus, eliminating the need to access the data from hard disk.

Ex. • Imagine your brain as your memory & a book as your hard drive

- Normally, when you need to recall something you have to go to the book, find the right page & read it. This takes time.
- But if you could store most of imp info directly in your brain (memory)? you could access it instantly!
- That is what in-memory analytics does for computers
- Instead of reading data from slow hard drive, in-memory analytics puts it in super fast RAM.
- This allows computer to analyze information much quicker, like finding patterns or making predictions

2) In-Database processing

- It is also called as in-database analytics.
- Typically the data from enterprise on line Transaction processing (OLTP) systems after cleaning up through the process of ETL is stored in EDW [Enterprise Data warehouse] or data marts.
- The huge datasets are then exported to analytical programs for complex & extensive computations.
- With in-database processing, the database pgm itself can run the computations eliminating the need for export & thereby saving on time.
- leading database vendors are offering this feature to large businesses.

3) Symmetric Multiprocessor System (SMP)

- In SMP, there is single common main memory that is shared by two or more identical processors
- Processor have full access to all I/O devices & are controlled by a single-operating system

3) Massively Parallel processing:

- It refers to the coordinated processing of programs by a no. of processors working parallel
- Processors each have their own OS & memory
- They work on different parts of the same program
- The MPP systems communicate using some sort of messaging interface
- The MPP systems are more difficult to program as the appn must be divided in such a way that all the executing segments can communicate with each other.

5) Different b/w Parallel & Distributed system

- Parallel database system is tightly coupled system
- The processors co-operate for query processing
- The user is unaware of the parallelism since he/she has no access to a specific processor of the system
- Either the processors have access to a common memory or make use of message passing for communication.

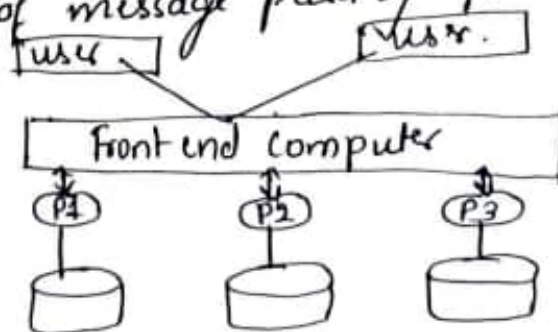


Fig: parallel system.

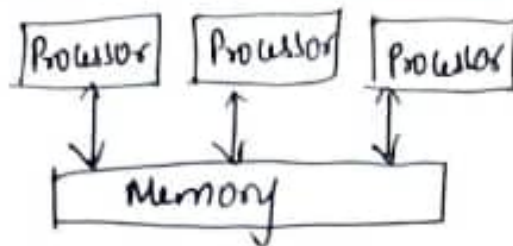


Fig: parallel system.

- Distributed Db systems are known to be loosely coupled & are composed by individual machines.
- Each of machine can run their individual appls & serve their own respective user.
- The data is usually distributed across several machines, thereby necessitating quite a no. of machines to be accessed to answer a user query.

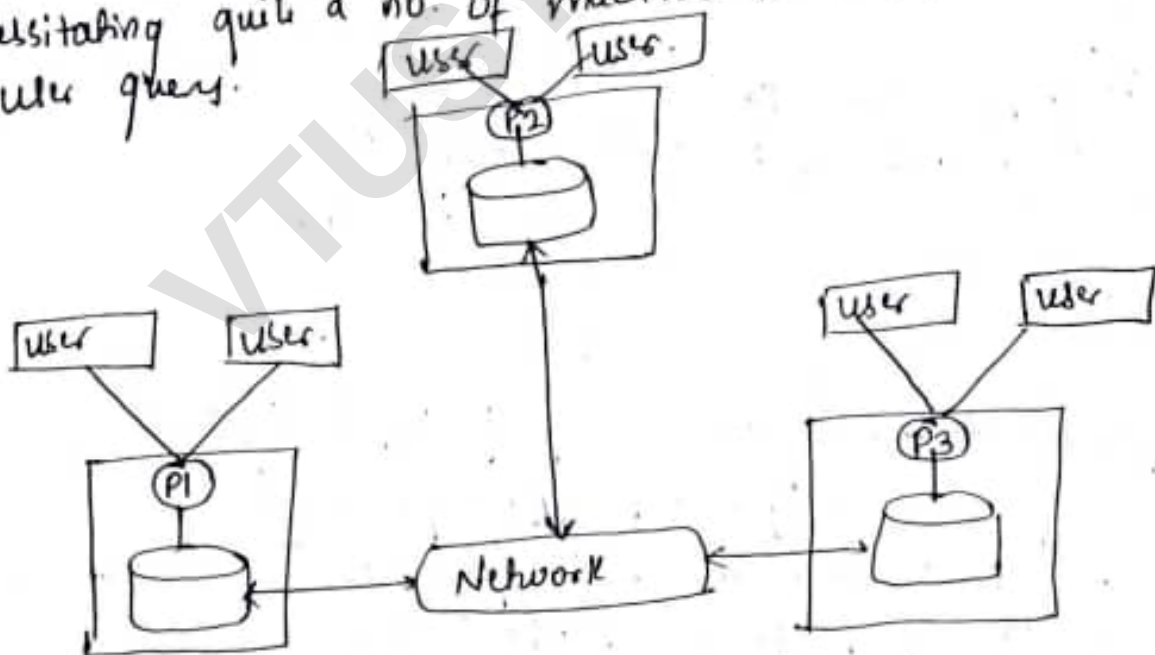


Fig: Distributed system.

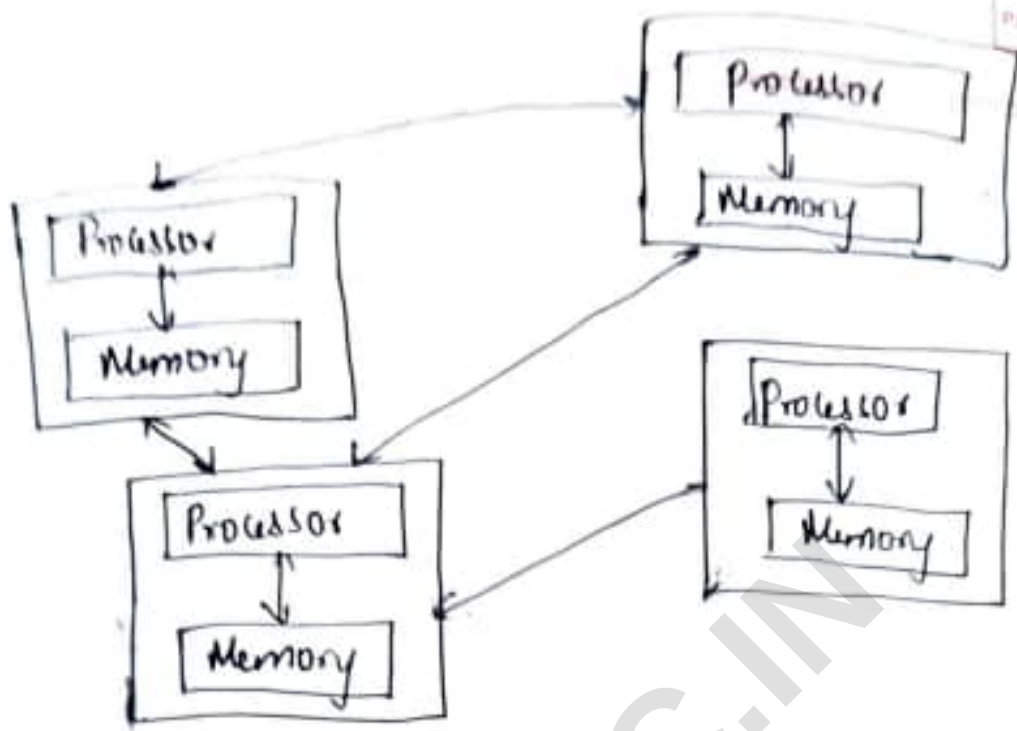


Fig: Distributed System.

Shared Nothing Architecture:

Three most common types of architecture for multiprocessor high transaction rate systems.

They are:

- 1) Shared Memory (SM)
- 2) Shared Disk (SD)
- 3) Shared Nothing (SN)

- In shared memory architecture, a common central memory is shared by multiple processors.
- In shared disk, multiple processors share a common collection of disks while having their own private memory.
- In shared nothing, neither memory nor disk is shared among multiple processors.

7) CAP Theorem Explained:

- The CAP theorem is also called as Brewer's Theorem.
- It states that in a distributed computing environment, it is impossible to provide the following guarantees.
 - 1) Consistency
 - 2) Availability
 - 3) Partition tolerance.

CAP Theorem

1. Consistency implies that every read fetches the last write
2. Availability implies that reads & writes always succeed. In other words, each non-failing node will return a response in a reasonable amount of time
3. Partition-tolerance implies that the system will continue to function when partitions occur.

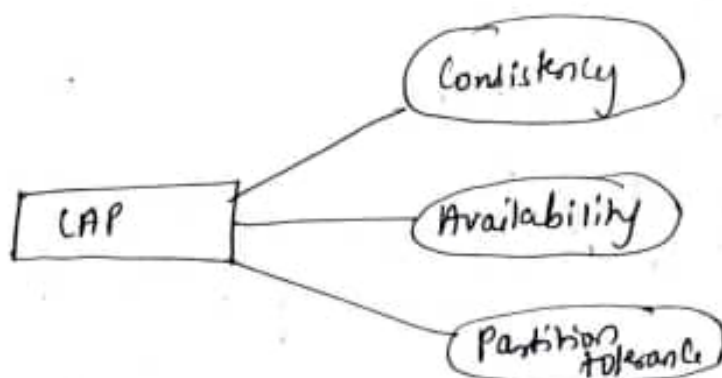
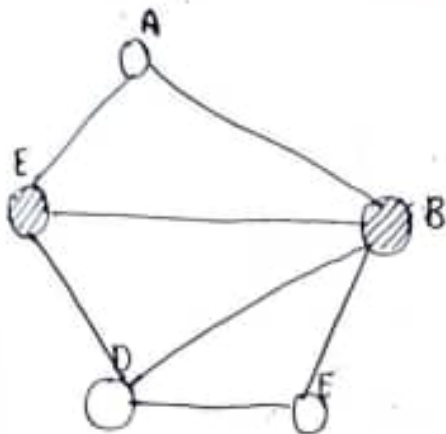


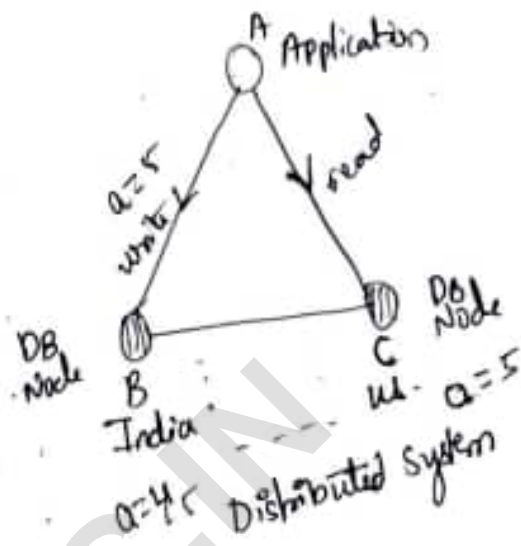
Fig: Brewer's CAP

Ex.

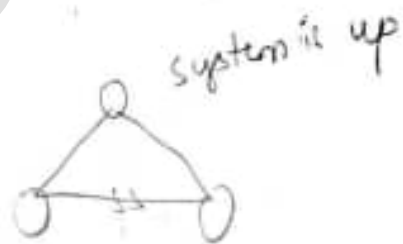


B, E are replica nodes.

Fig: Distributed System.



A - All nodes should respond.



4.1 NOSQL (NOT ONLY SQL)

→ The term was first coined by Carlo Strozzi in 1998 to name his lightweight, open-source, relational database that did not expose the standard SQL interface.

→ Few features of NoSQL database as follows:

- 1) They are open source.
- 2) They are non-relational.
- 3) They are distributed.
- 4) They are schema-less.
- 5) They are cluster friendly.
- 6) They are born out of 21st Century web applications.

4.1.1. Where it is used?

→ They are widely used in big data & other real-time web applications.

Scale out or horizontal scaling: Increasing System Capacity by adding more machines to distribute the workload.

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Scaling up or vertical scaling: upgrading the existing machine.

- It is used to store log data which can then be pulled for analysis.
- Likewise it is used to store social media data & all such data which cannot be stored & analyzed comfortably in RDBMS.

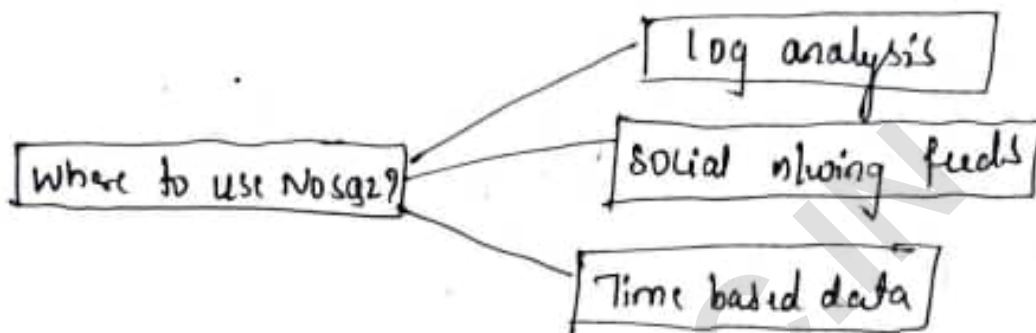


Fig: Where to use NoSQL?

4.1.2 What is it?

- NoSQL stands for Not Only SQL.
- These are non-relational, open source, distributed databases.
- They are highly popular today owing to their ability to scale out or scale horizontally & dealing with a rich variety of data; structured, semi or unstructured.
- Below fig shows additional features of NoSQL.

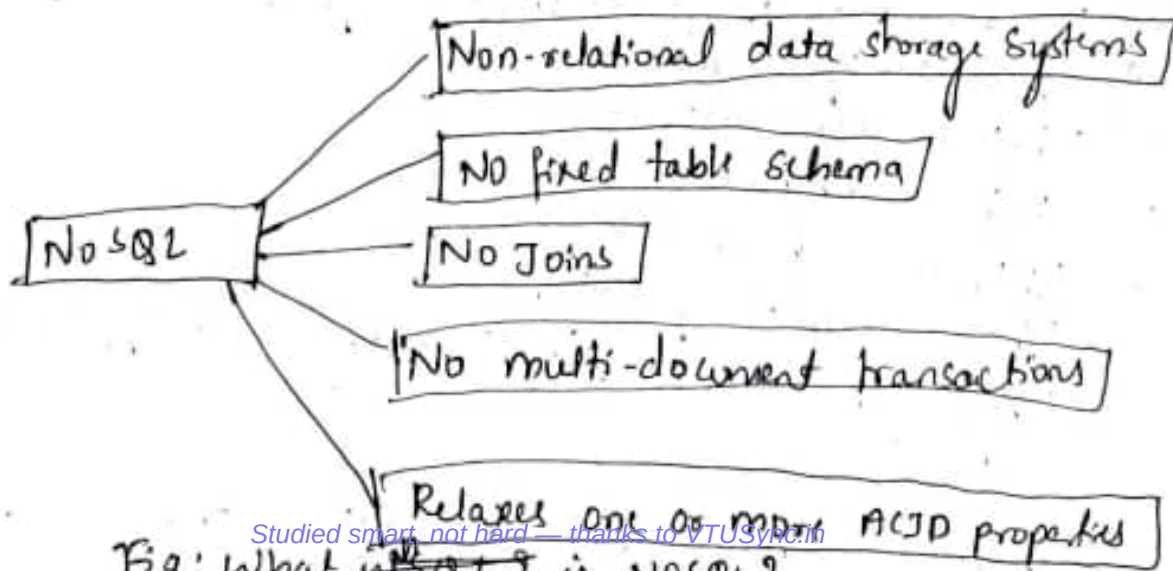


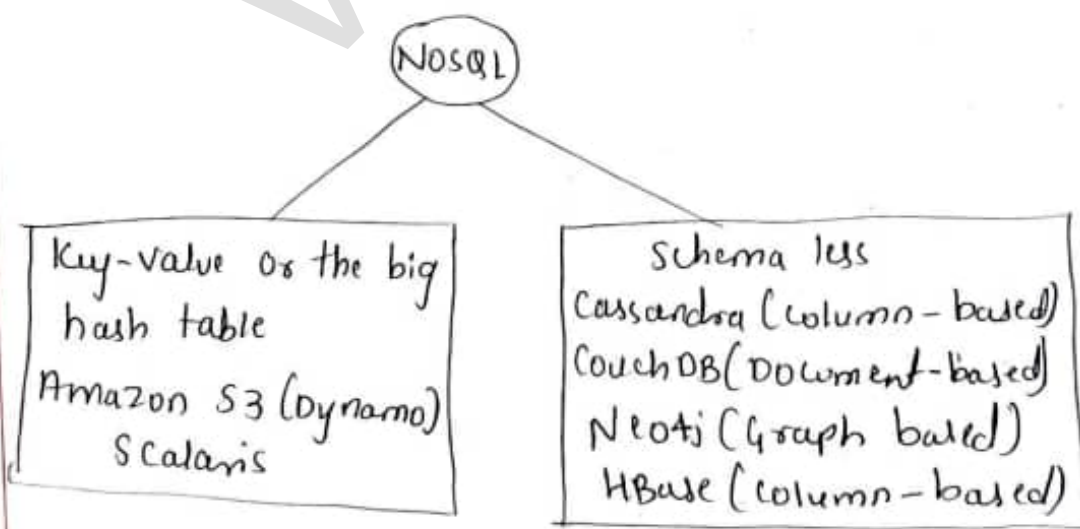
Fig: What is it? is NoSQL?

1. Are non-relational: They do not adhere to relational data model. In fact they are either key-value pairs or document-oriented.
2. Are distributed:
 - The data is distributed across several nodes in a cluster constituted of low-cost commodity h/w.
3. Offer no support for ACID properties
 - They do not offer support ACID properties of transactions
4. Provide no fixed table schema:
 - NoSQL databases are becoming increasingly popular owing to their support for flexibility to the schema.

4.1.3 Types of NoSQL Databases.

NoSQL can be broadly classified into

1. Key value or the big hash table
2. Schema-less



- 1) Key-value: It maintains a big hash table of keys & values

Ex:

Key	Value	Ex:	Value
First Name	Simmonds	Keys	{ "name": "Michael", "Age": 123 }
Last Name	David	123	

2. Document:

→ It maintains data in collections constituted of documents

Ex: MongoDB, Apache Couch DB, Couchbase etc.

"Book Name": "Fundamentals of Business Analytics",

"Publisher": "Wiley India",

"Year of publication": "2011"

Customer ID	101	102	103	104
Name	John	Mary	Bob	Alice
Age	30	25	35	28
City	IND	CHI	NY	USA

3. Column

→ Each storage block has data from only one column.

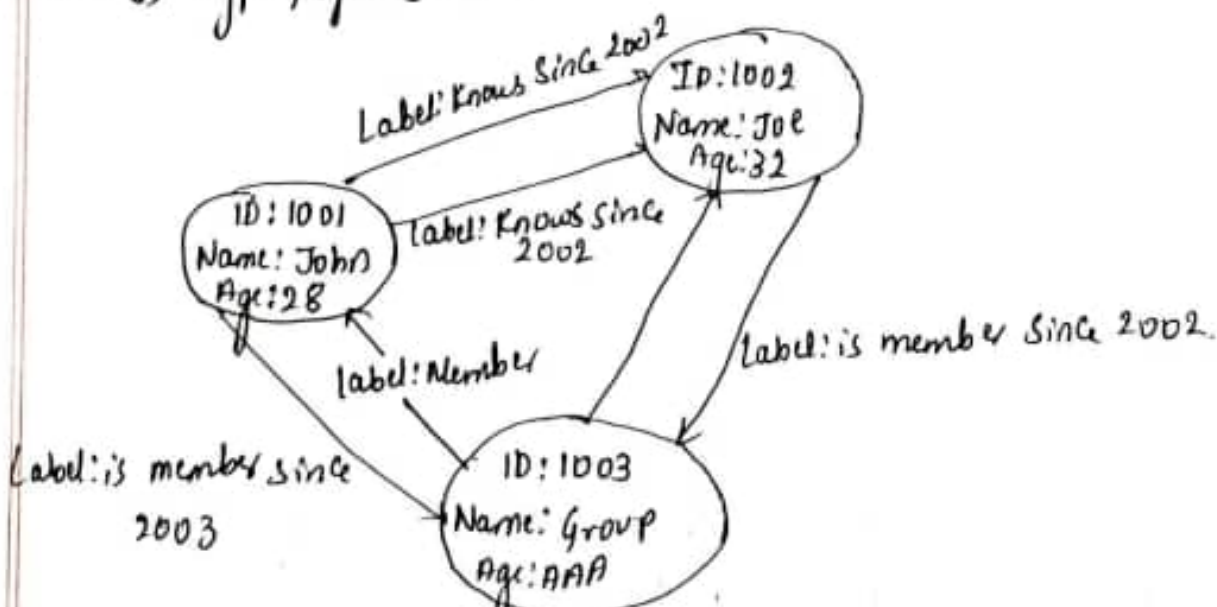
Ex: Cassandra, HBase etc.

4. Graph:

→ They are also called n/w database.

→ A graph stores data in nodes

Ex: Neo4j, HypergraphDB etc.



4.14 Why NOSQL?

- It has scale out architecture instead of the monolithic architecture of relational databases.
- It can house large volumes of structured, semi-structured & unstructured data.
- Dynamic schema:
NOSQL database allows insertion of data without a pre-defined schema.
- Auto-sharding:
It automatically spreads data across an arbitrary no. of servers.
- Replication:
It offers good support for replication which in turn guarantees high availability, fault tolerance & disaster recovery.

4.15 Advantages of NOSQL

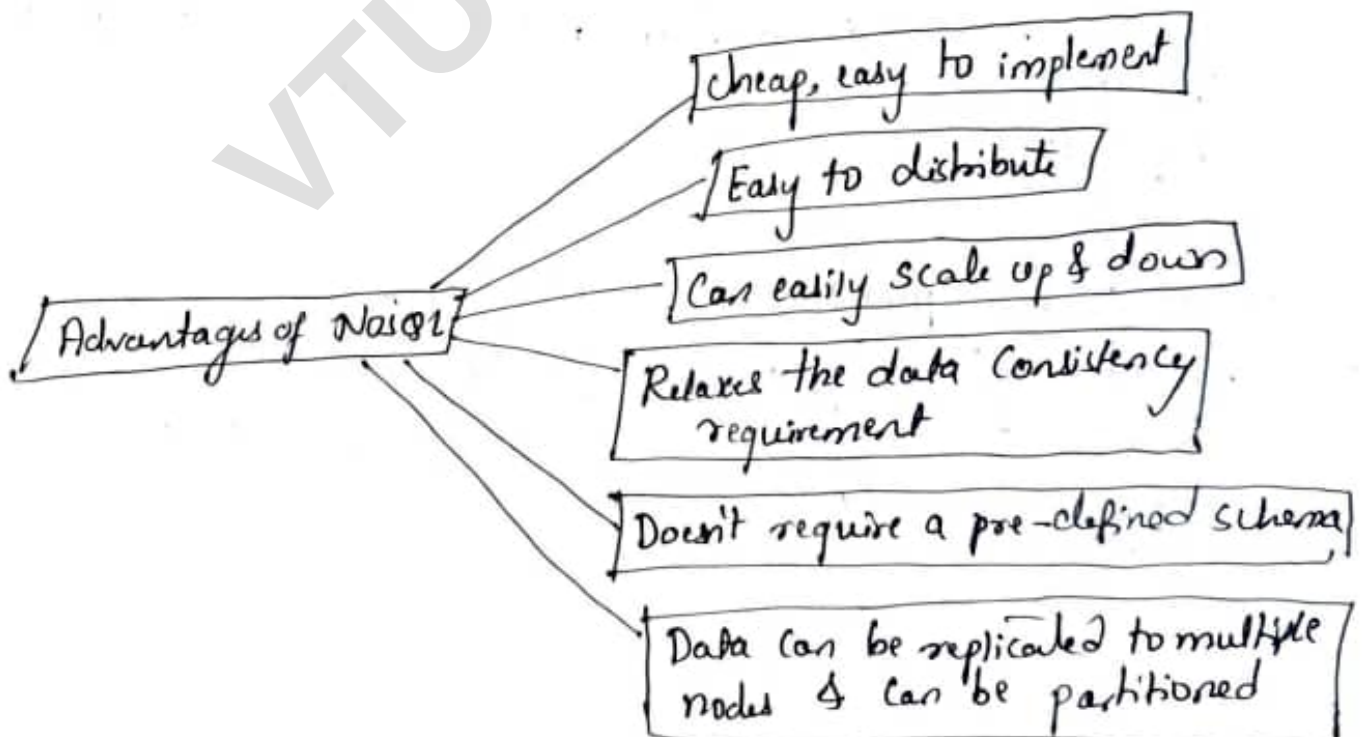


Fig: Advantages of NOSQL.

1. Can easily scale up & down: NoSQL DB ~~allows~~ supports scaling rapidly & elastically & even allows to scale to the cloud.

2. Don't require a predefined schema:

- Not require any adherence to predefined schema.
- Ex:

{-id:101, "BookName": "Fundamentals of Business Analytics", "AuthorName":

"Seema Ahanya", "publishes": "Wiley India"}

{-id:102, "BookName": "Big Data & Analytics"}

3. Cheap, easy to implement:

Deploying properly allows the benefits of scale, high availability, fault tolerance, etc while lowering operational costs.

4. Relax data consistency requirement:

No SQL databases have adherence to CAP theorem.

5. Data can be replicated to multiple nodes & can be partitioned

a) Sharding

b) Replication

4.1.7 Use of NoSQL in Industry

→ NoSQL is being put to use in varied industries

→ They are used to support analysis for applications such as web user data analysis, log analysis, sensor feed analysis, making recommendations for upsell & cross-sell

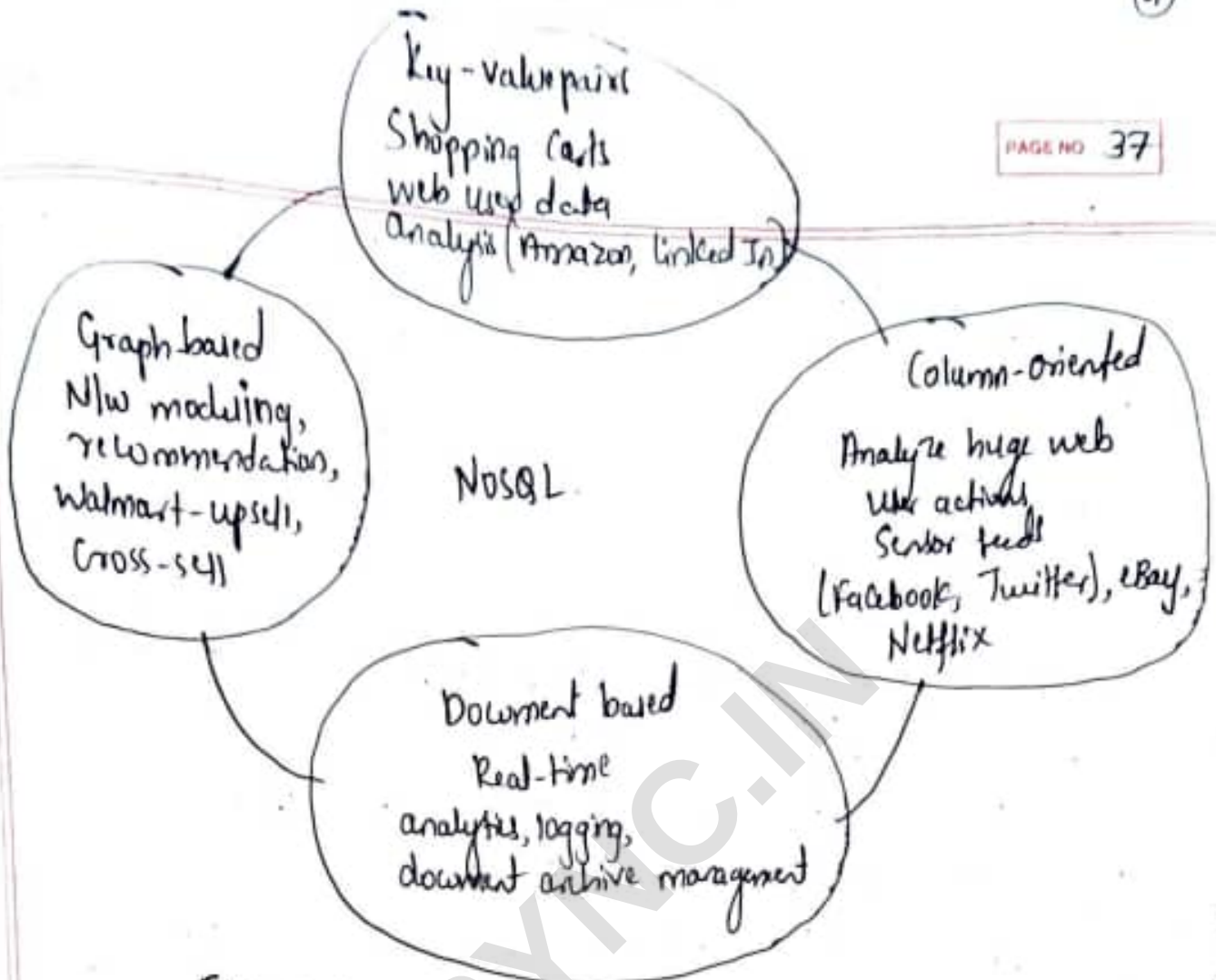


Fig: Use of NoSQL in industry.

4.1.9 SQL Versus NoSQL

SQL

- 1) Relational database
- 2) Relational model
- 3) Pre-defined Schema
- 4) Table based databases
- 5) Vertically Scalable
- 6) Use SQL
- 7) Not preferred for large datasets
- 8) Not a best fit for hierarchical data

NOSQL

- 1) Non-relational, distributed database
- 2) Model-less approach
- 3) Dynamic-schema for unstructured data
- 4) Document-based or graph based or wide column store or key-value pair databases
- 5) Horizontally Scalable
- 6) Uses Unql (unstructured Query language)
- 7) Largely preferred for large datasets
- 8) Best fit for hierarchical storage as it follows the key-value pair of storing data similar to JSON

SQL

- 9) Emphasis on ACID properties
- 10) Excellent support from vendors
- 11) Ex: Oracle, DB2, MySQL, MSSQL, PostgreSQL, etc.

NoSQL

- 9) Follows Brewer's CAP theorem
- 10) Relies heavily on community support
- 11) MongoDB, HBase, Cassandra, Redis, Neo4j, CouchDB, Riak etc.

4.1.10 NewSQL: It is combination of both SQL & NoSQL

- NewSQL is a unique database system that combines ACID properties Compliance with horizontal scaling
- The database system strives to keep the best of both worlds
- This new modern RDBMS is called NewSQL
- It supports relational data model & uses SQL as their primary interface

CHARACTERISTICS OF NewSQL

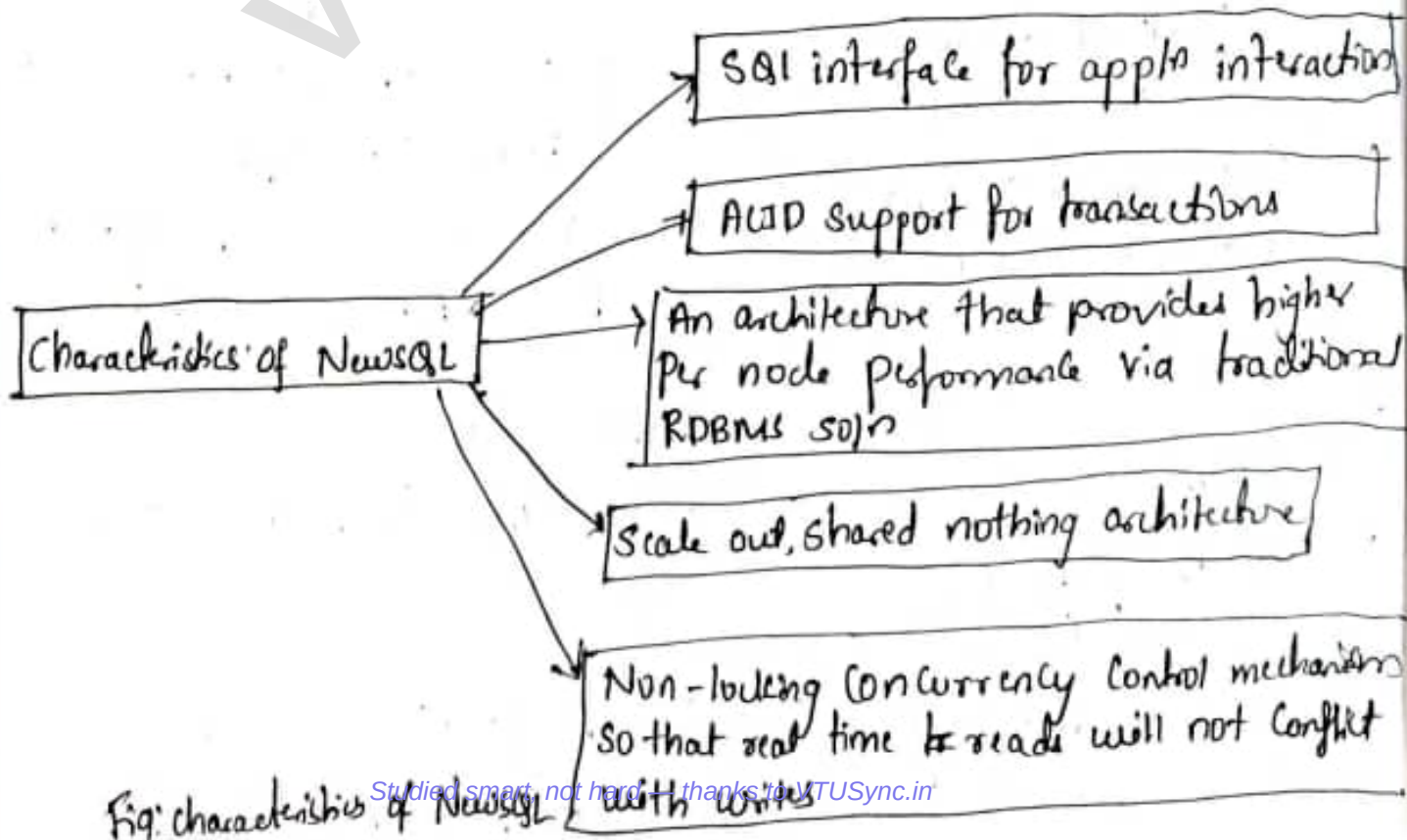


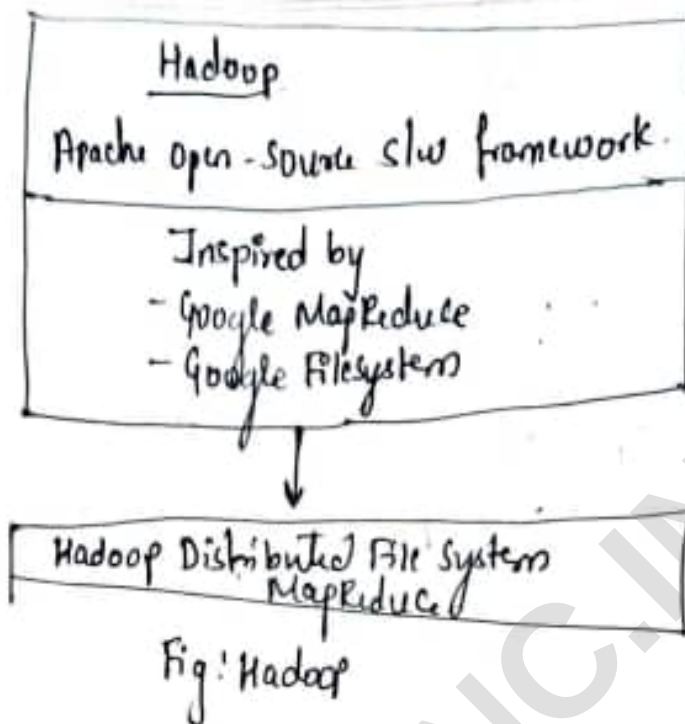
Fig: characteristics of NewSQL

Comparison of SQL, NoSQL & NewSQL

	SQL	NoSQL	NewSQL
1. Adherence to ACID properties	Yes	No	Yes
2. OLTP/OLAP	Yes	No	Yes
3. Schema rigidity	Yes	No	Maybe
Adherence to data model	Adherence to relational model	No	Maybe
4. Data Format Flexibility	No	Yes	Maybe
5. Scalability	Scale up Vertical scaling	Scale out horizontal Scaling	Scale out
6. Distributed Computing	Yes	Yes	Yes
7. Community Support	Huge	growing	Slowly growing

4.2 Hadoop

- Hadoop is a framework that allows for distributed processing of large datasets across clusters of commodity computers using a simple programming model.
- It is an open source project by Apache Software Foundation
- It has ability to store & analyze data present in diff machines at diff location very quickly & in a very cost effective manner.
- Hadoop was started with Doug Cutting & Mike Cafarella, in the year 2002 when they both started to work on Apache Nutch project
- ~~Approximate cost half~~ This framework is basically written in java.
- Hadoop uses Google's MapReduce & Google File System technologies as its foundation.



4.2.1 Features of Hadoop

- It is optimized to handle massive quantities of structured, semi-structured & unstructured data, using commodity h/w, that is relatively inexpensive (readily available, & inexpensive computing components) Computers.
- Hadoop has a shared nothing architecture.
- It replicates data across multiple computers so that if one goes down, the data can still be processed from another machine that stores its replica.
- Hadoop is for high throughput rather than low latency. It is batch operation handling massive quantities of data; therefore the response time is not immediate.
- It complements OLTP & OLAP. However, it is not a replacement for a relational database mgmt system.
- It is not good when work cannot be parallelized or when there are dependencies with the data.
- It is not good for processing small files. It works best with huge data files & datasets.

Key Advantages of Hadoop

(41)

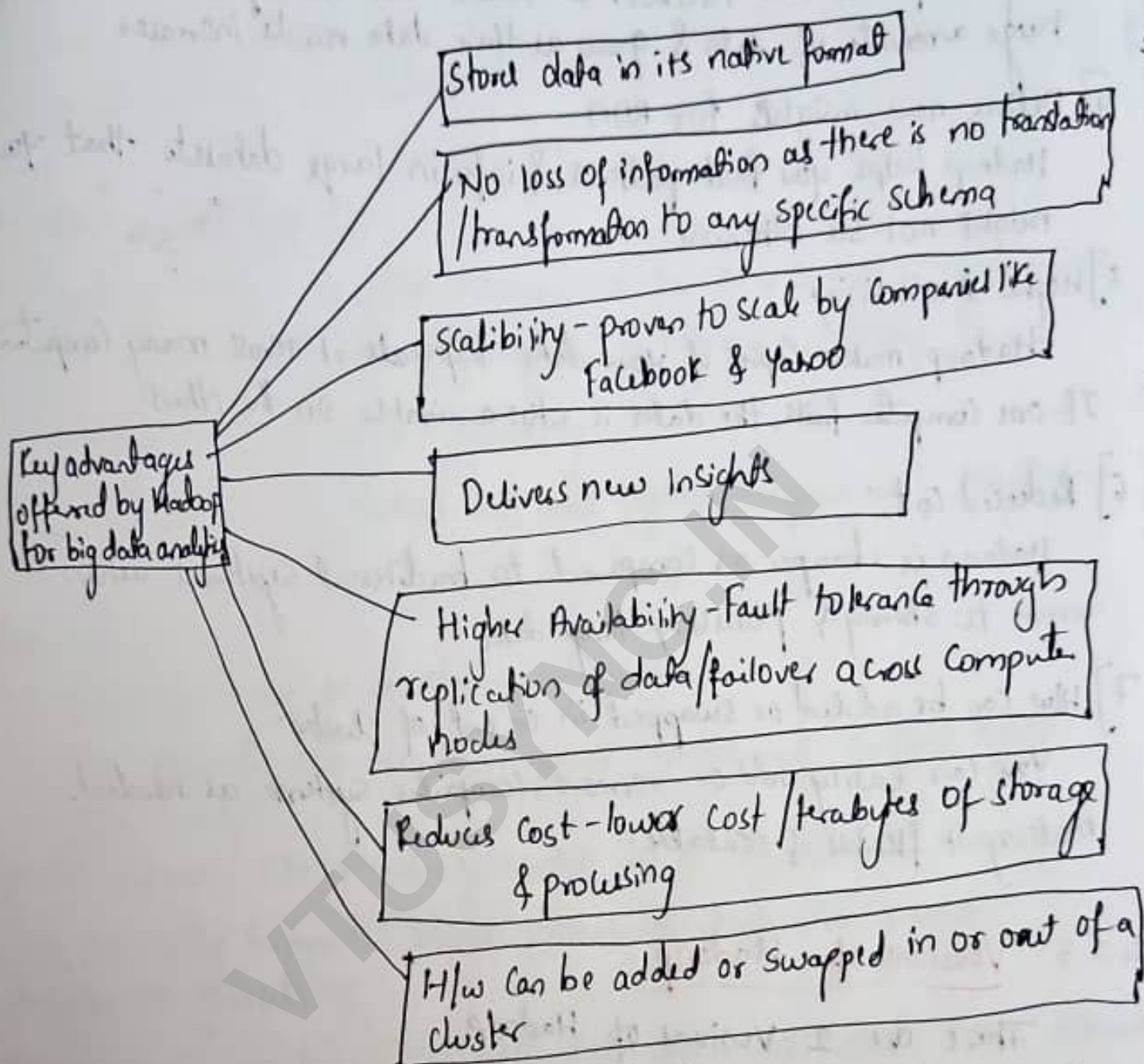


Fig: Key Advantages of Hadoop

1] Stores data in its native format:-

- Hadoop doesn't force you to change your data into a specific format before storing it.
- You can store it as it is, which saves time & effort.

2] No loss of information as there is no translation to any specific schema:
→ As there is no change in the data format, so we don't lose any of the original details or info.

3] Scalable:

Big Companies like Facebook & Yahoo use Hadoop bcz it can handle huge amounts of data & grow as their data needs increase

4] Derive new insights for BOA

Hadoop helps you find patterns & info in large datasets that you might not see otherwise

5] Higher Availability:

Hadoop makes copies of your data & spreads it across many computers. If one computer fails, the data is still available on the others

6] Reduced cost:

Hadoop is cheaper as compared to traditional systems when it comes to storing & processing large data

7] How can be added or swapped in or out of cluster:

You can easily add or remove computer systems as needed, making it flexible & scalable

4.2.3 Versions of Hadoop

There are 2 Versions of Hadoop

1) Hadoop 1.0

2) Hadoop 2.0

Hadoop 1.0
MapReduce (cluster Resource Manager & Data Processing)
HDFS (Redundant, reliable storage)

Hadoop 2.0	
MapReduce (Data processing)	others (Data processing)
YARN (cluster Resource Manager)	
HDFS (Redundant, reliable storage)	

Fig: Versions of Hadoop

* Hadoop 1.0

It has two main parts

- 1) Data storage framework
- 2) Data processing framework

Limitations of Hadoop 1.0

- 1) The first limitation was the requirement for MapReduce programming expertise along with proficiency required in other PL, notably Java.
- 2) It supported only batch processing
- 3) Major limitation was Hadoop 1.0 was tightly coupled with MapReduce.

* Hadoop 2.0:

- It continues to be data storage framework.
- A new separate resource management framework called YARN has been added.
- YARN enhances the flexibility, scalability & efficiency of the appls.
- It not only supports batch processing but also supports real-time processing.
- MapReduce is no longer the only data processing option; other alternate data processing function such as data standardization, master data management can be performed natively in HDFS.

4.2.4 Overview of Hadoop Ecosystems

These are components available in the Hadoop ecosystem for data ingestion, processing & analysis.

Data ingestion → Data processing → Data Analysis
 (Process of collecting & importing data from various sources into a central location)
 Components that help with Data Ingestion are

1. Sqoop
2. Flume

Components that help with Data processing are

1. MapReduce
2. Spark

Components that help with Data Analysis are

1. Pig
2. Hive
3. Impala

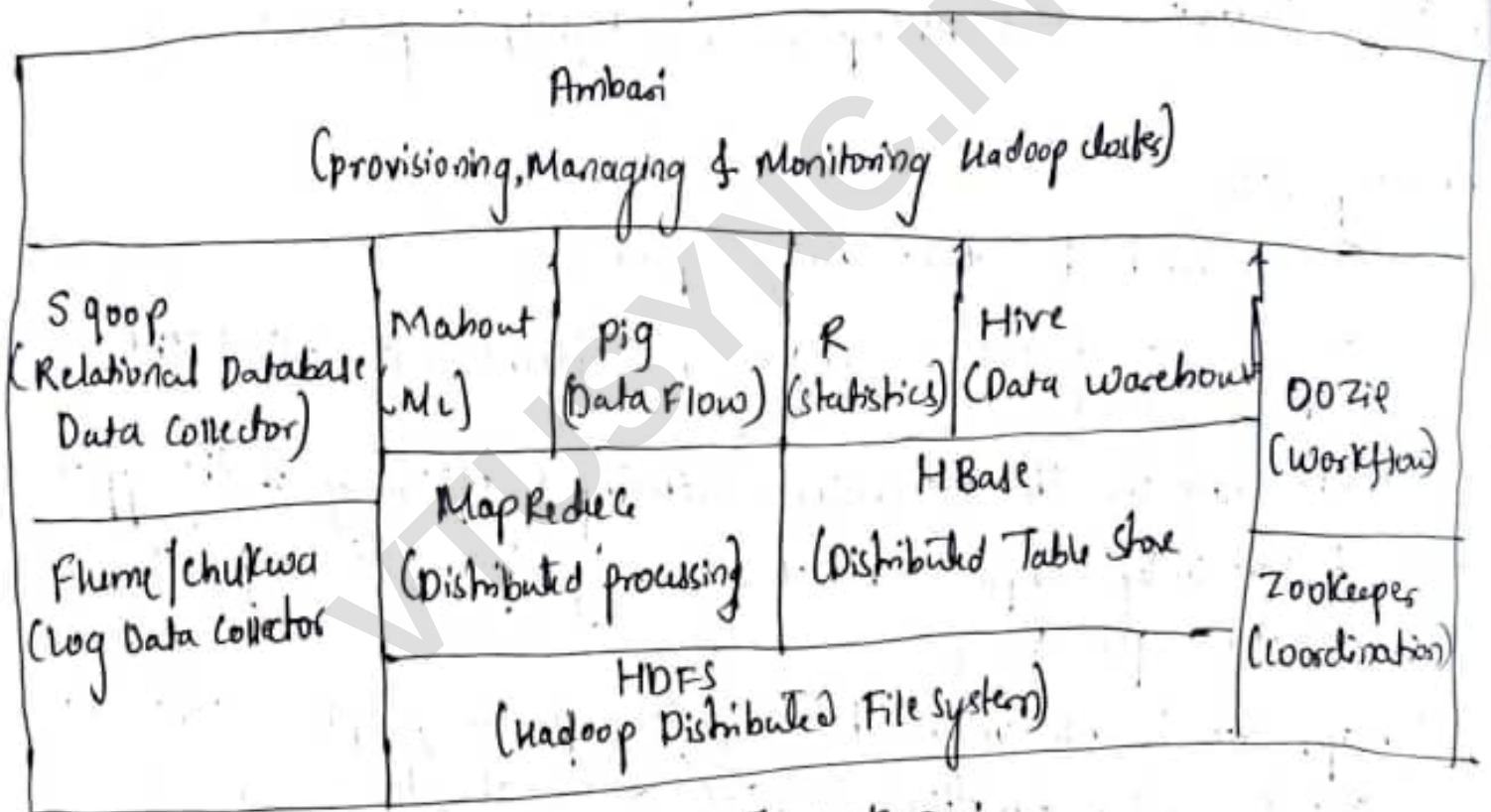


Fig: Hadoop Ecosystem

1) HDFS: It is the foundation of Hadoop ecosystem designed to store & manage large datasets across a cluster of commodity h/w

How it works:

- Data is broken down into blocks & distributed across multiple nodes in the cluster
- Replication is used to ensure fault tolerance
- HDFS is optimized for sequential read/write operations making

④ ideal for batch processing

note 1

(us)

2) MapReduce:

→ ~~This is the~~ This is the programming model & processing engine for Hadoop.

→ It allows for 11^{th} processing of large datasets

• How it works:

- "Map" phase processes input data & generates key-value pairs
- "Reduce" phase aggregates & combines the results for the "map" phase to produce the final output
- MapReduce jobs are executed across the nodes in the Hadoop clusters, enabling 11^{th} processing

3) YARN (Yet Another Resource Negotiator):

- This is the resource management layer of Hadoop.
- It manages cluster resources & schedules apps.

• How it works:

- It separates resource management from job scheduling
- The Resource Manager allocates resources to applications, & the Application Master manages the execution of individual jobs.
- YARN allows multiple apps to run on the Hadoop clusters, improving resource utilization

4) Data Ingestion Tools: [Flume / Chukwa]:

→ These tools are used to collect & ingest streaming data into HDFS

How it works

- Flume is used to collect & transport large volume of log data
- Chukwa is also used for log collection & monitoring
- They can handle data from various sources & reliably deliver it to HDFS

5) Additional tools

(i) HBase (Distributed Table store):

→ A NoSQL, distributed, column-oriented database built on top of HDFS

→ Provides real-time read/write access to large datasets

(ii) Hive (Data warehouse):

→ A data warehousing system that allows querying & analyzing data in HDFS using SQL-like queries (HiveSQL)

(iii) Pig (Flow):

→ A high level platform for creating MapReduce programs using scripting language (Pig Latin)

(iv) Mahout (ML):

→ A library of Scalable machine learning algo that run on Hadoop

(v) Sqoop (Relational DB):

→ A tool to transfer data b/w Hadoop & relational DB

(vi) Ambari (provisioning, Managing & monitoring Hadoop clusters):

→ A web based tool for managing & monitoring Hadoop clusters

(vii) Zookeeper (Coordination):

→ A distributed coordination service that provides synchronization, configuration mgmt & naming services for distributed apps

(viii) Oozie (Workflow):

→ A workflow scheduler system to manage Hadoop jobs

4.2.5 Hadoop Distributions:

→ Hadoop is an open-source Apache project. It

→ The core aspects of Hadoop include

1) Hadoop Common

2) HDFS

3) Hadoop YARN (Yet Another Resource Negotiator)

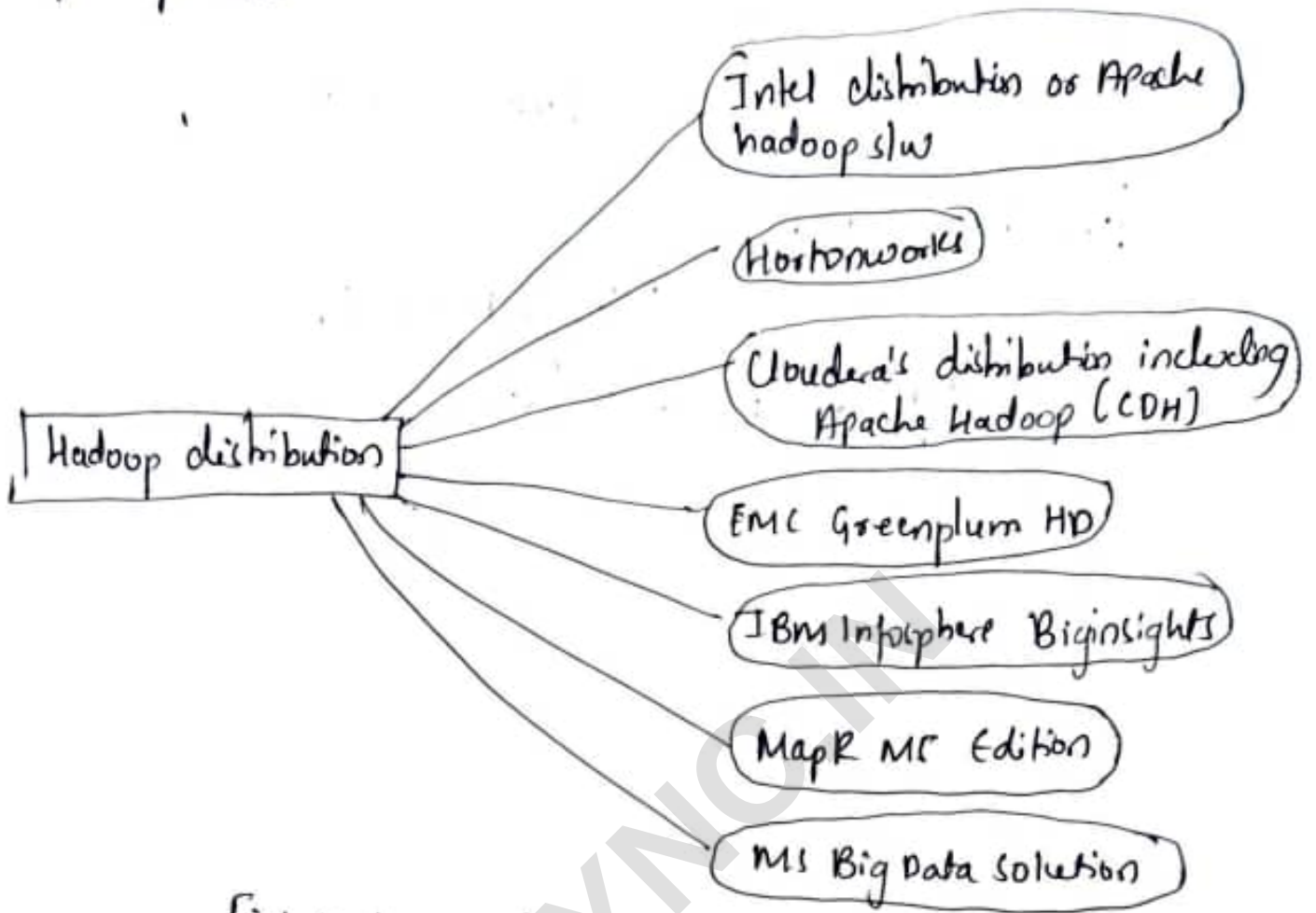


Fig: Hadoop Distributions.

4.2.6 Hadoop VS SQL

Hadoop	SQL
Scale out	Scale up
Key-value pairs	Relational table
Functional programming	Declarative queries
Offline batch processing	Online transaction processing

4.2.7 Integrated Hadoop System offered by leading Market Vendors.

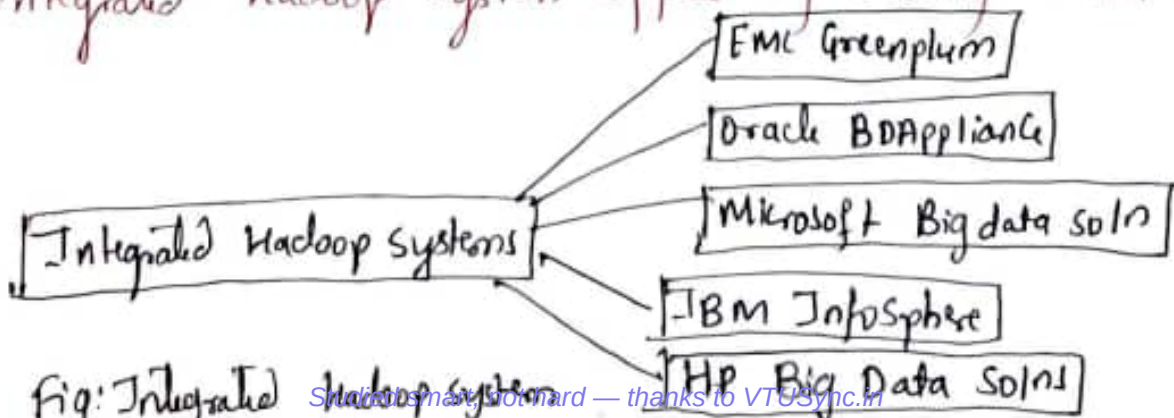


Fig: Integrated Hadoop system

4.2.8 Cloud-Based Hadoop solutions

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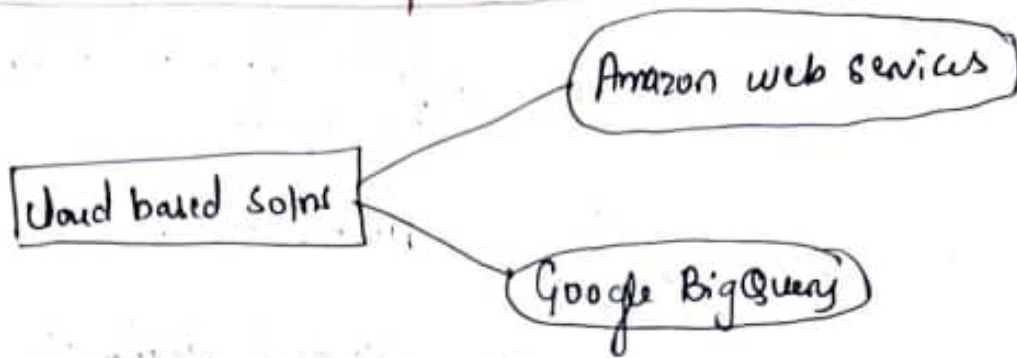


Fig: cloud Based solutions.

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